

Lesson 1: Getting Started with Tableau and Building Basic Visualizations

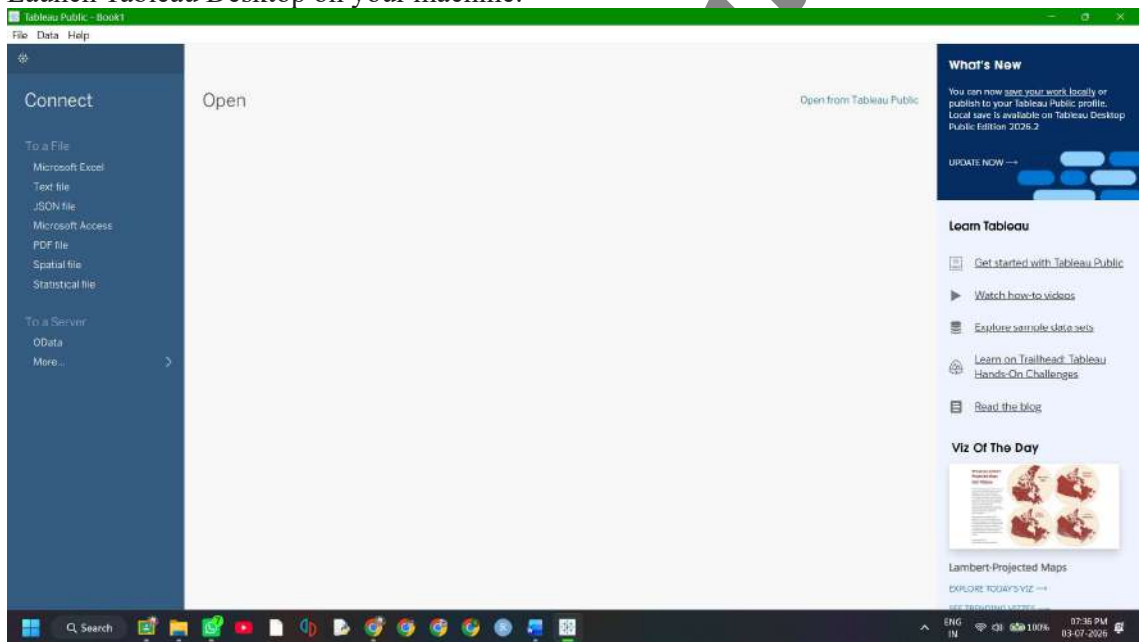
- a) Connect to a CSV dataset and explore the Tableau interface
- b) Identify and differentiate Measures and Dimensions
- c) Convert fields between discrete and continuous and observe changes
- d) Create and customize a basic bar chart
- e) Build a line chart for time-series analysis
- f) Create a geographic map visualization
- g) Combine multiple charts into a simple interactive dashboard

Dataset: Superstore.

kaggle link: <https://www.kaggle.com/code/rajatraj0502/superstore-dataset>

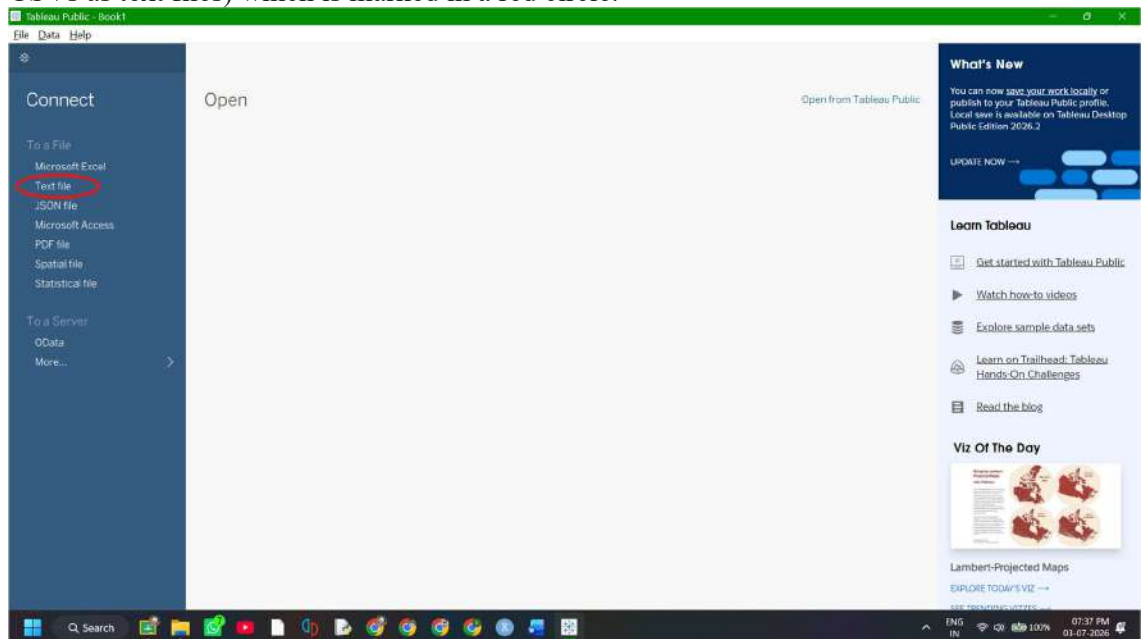
Part A: Connect to Dataset & Explore the Interface

1. Launch Tableau Desktop on your machine.

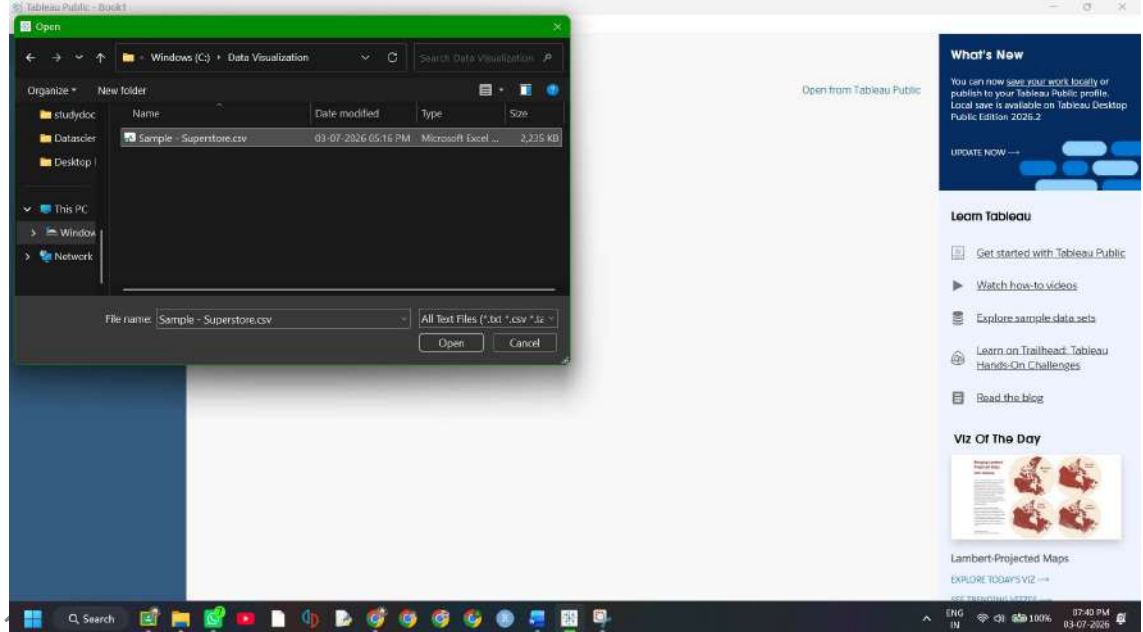


DATA VISUALIZATION WITH POWER BI AND TABLEAU

2. On the left-hand blue navigation pane under Connect, click on Text File (Tableau treats CSVs as text files) which is marked in a red circle.

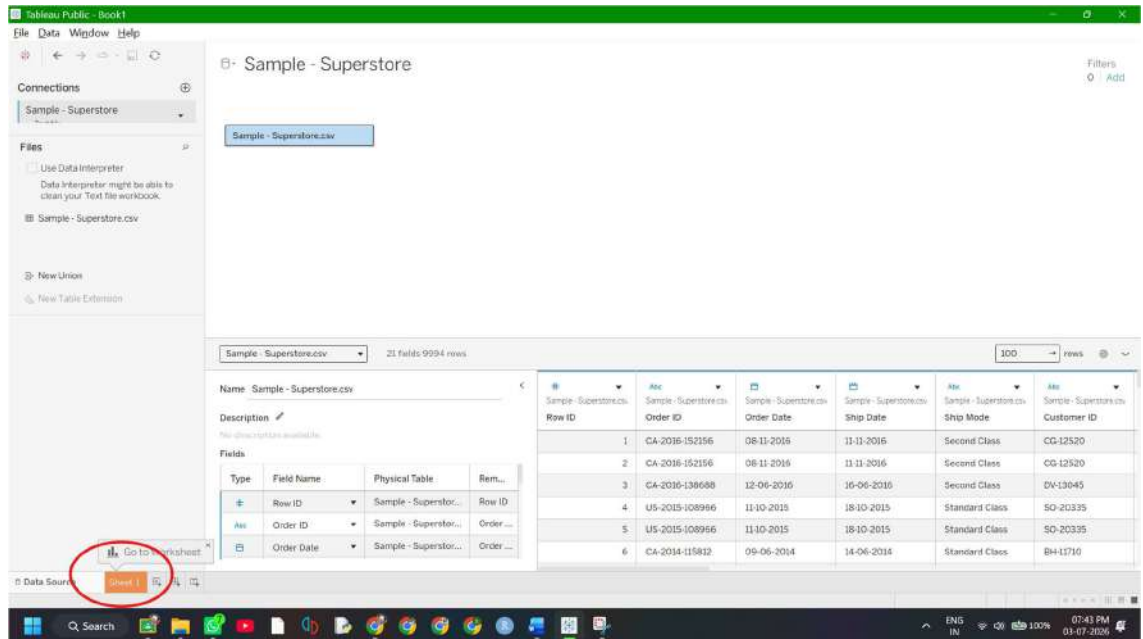


3. Select your CSV file from your Mac Finder and click Open.



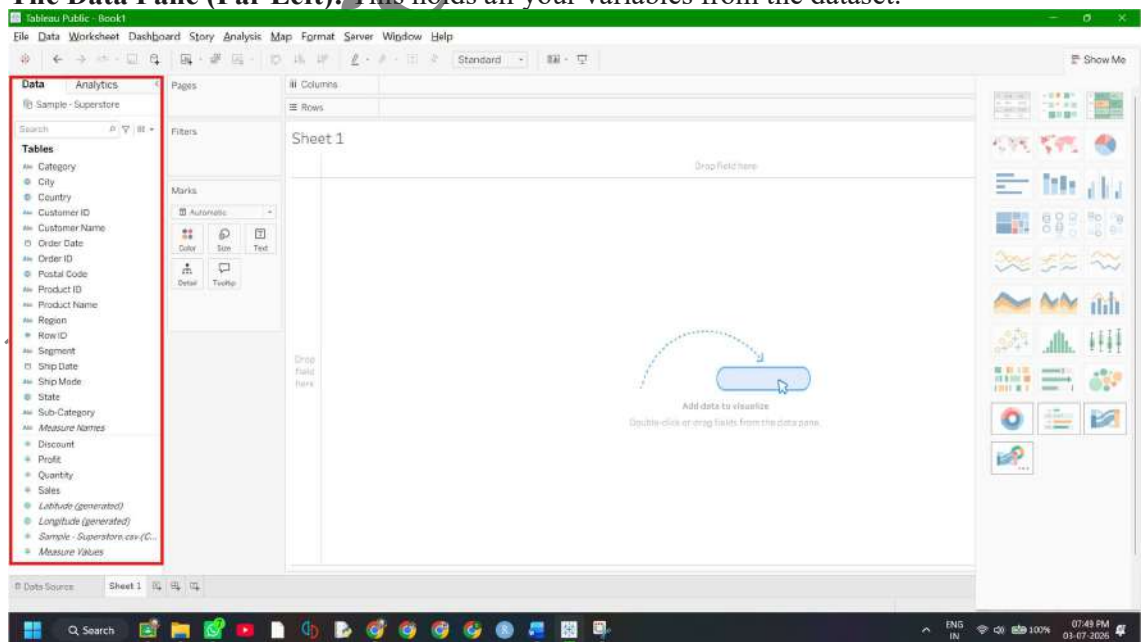
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- You will be taken to the Data Source page. Here, preview your data rows at the bottom to ensure they look correct, then click Sheet 1 at the bottom left to enter your workspace marked in a red circle inside the snapshot. Tableau will bypass its faulty drivers, create a native workspace called Clipboard, and instantly open up **Sheet**.



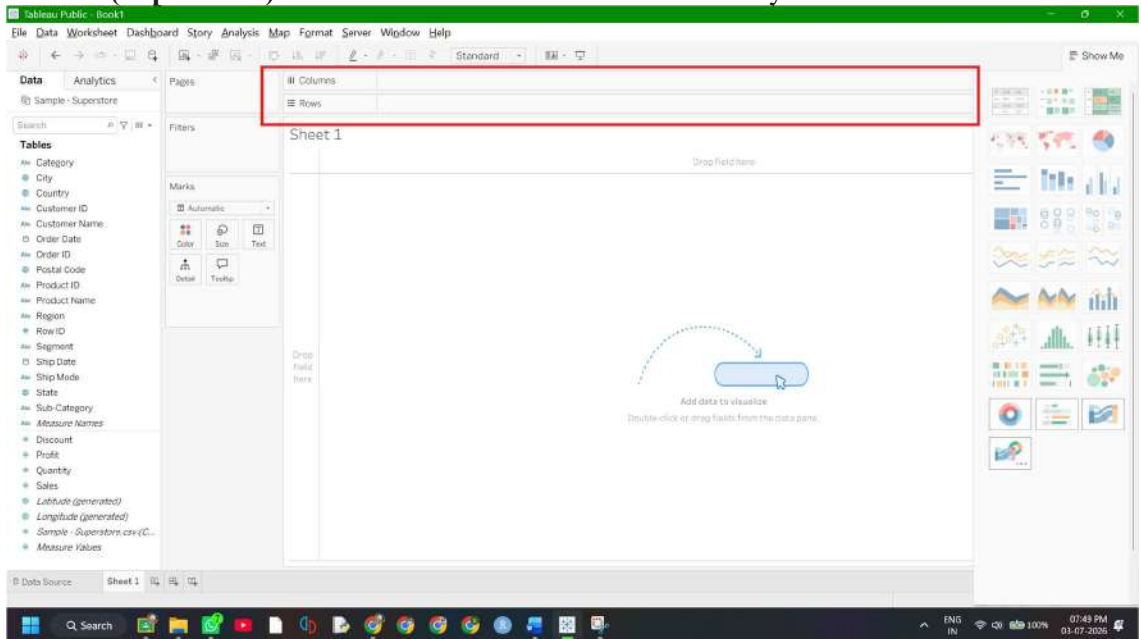
Navigating the Interface

- The Data Pane (Far Left):** This holds all your variables from the dataset.

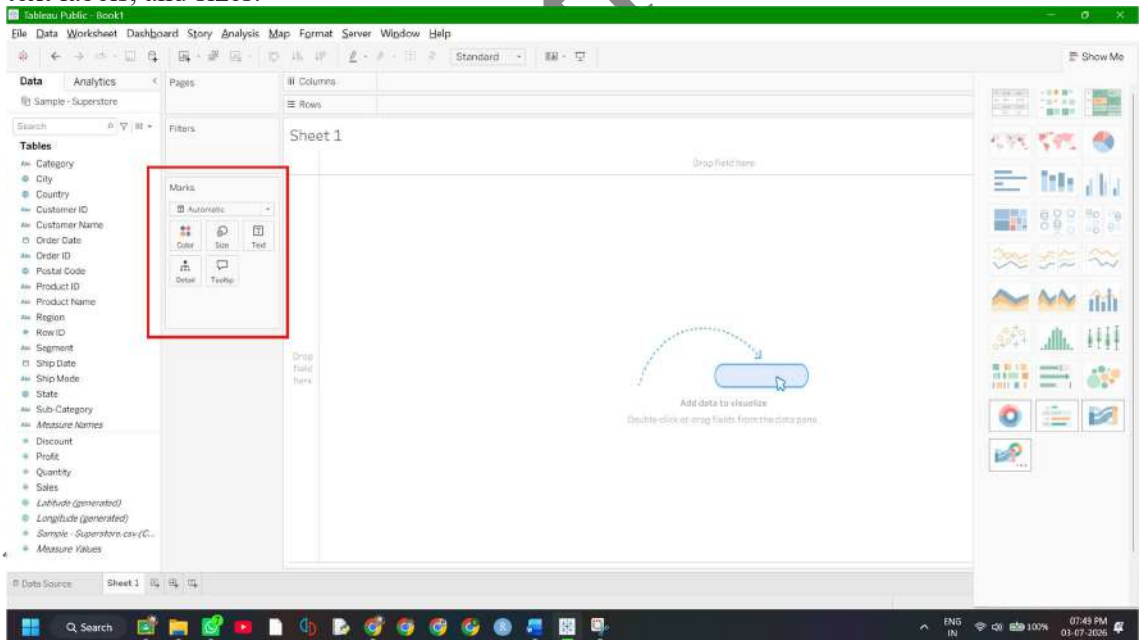


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- **Shelves (Top/Center):** The **Columns** and **Rows** shelves act as your X and Y coordinates.

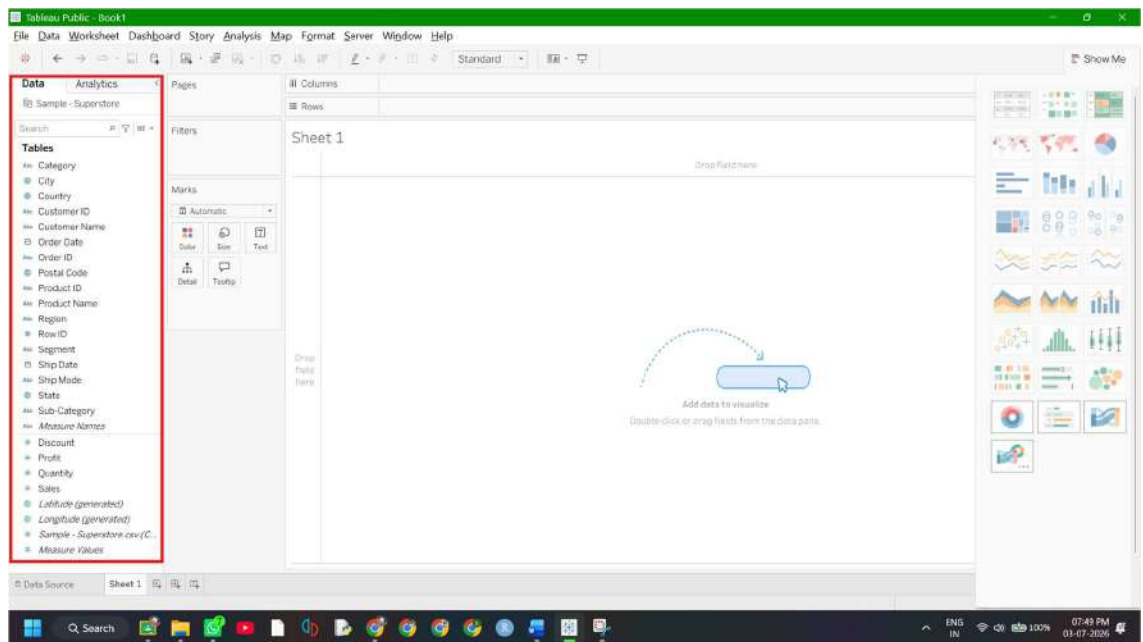


- **The Marks Card (Left of center):** This controls your visual design properties like color, text labels, and sizes.

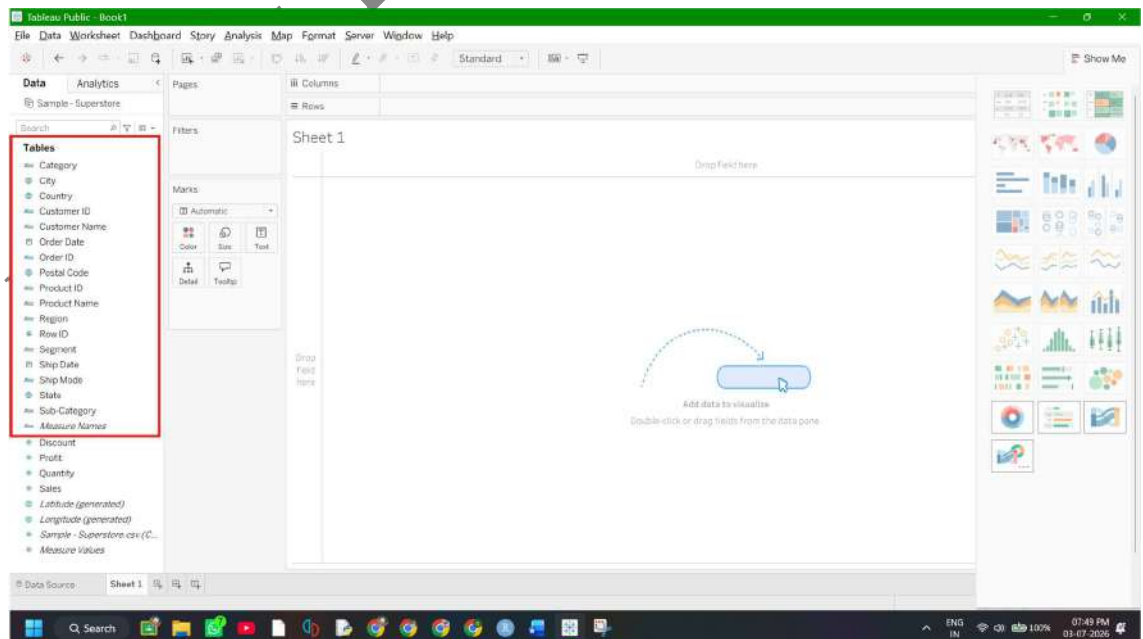


Part B: Identify and Differentiate Measures and Dimensions

Look closely at your **Data Pane** on the left. Tableau reads the metadata of the Superstore CSV and splits your items automatically:

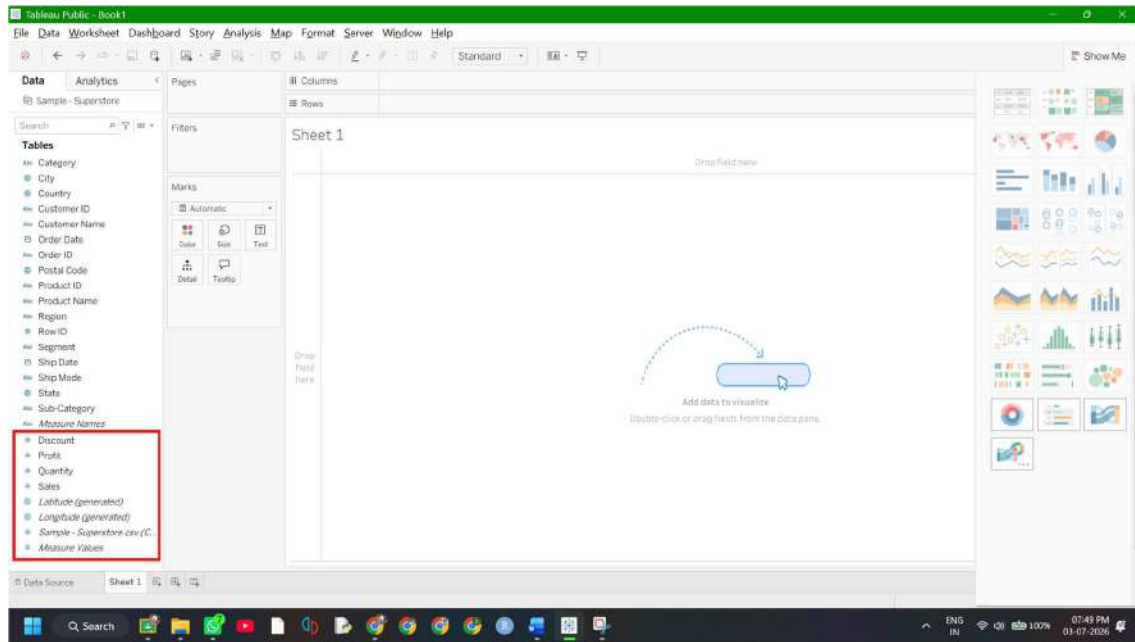


- **Dimensions (Categorical / Blue Icons):** Fields containing text, dates, or geography. In Superstore, look for fields like Category, Segment, State, or Order Date. They slice your data into distinct groups.



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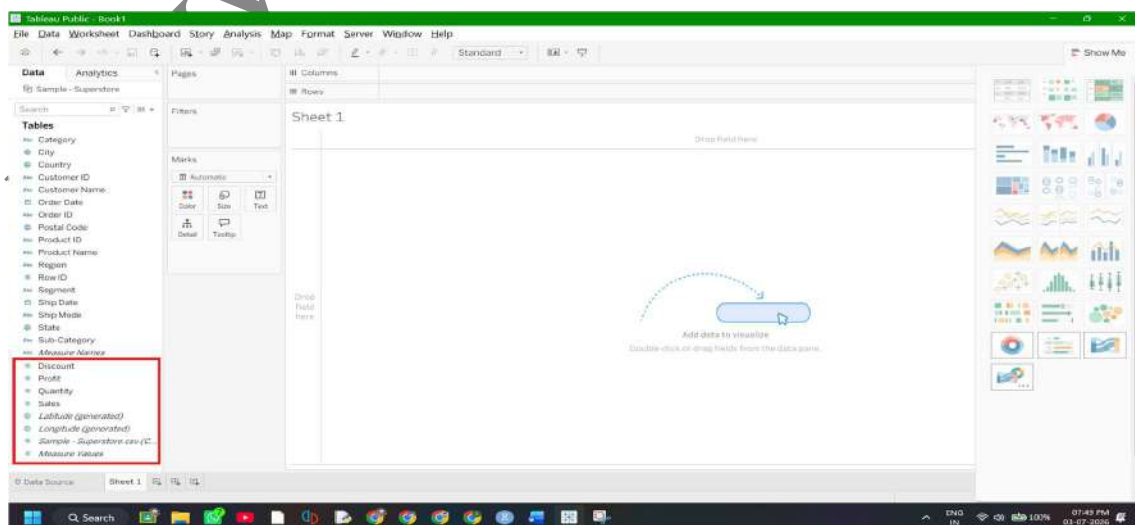
- **Measures (Numerical / Green Icons):** Fields containing raw quantitative values. Look for fields like Sales, Profit, or Quantity. Tableau will automatically summarize or average these values whenever you place them on a chart.



Part C: Convert Fields Between Discrete and Continuous

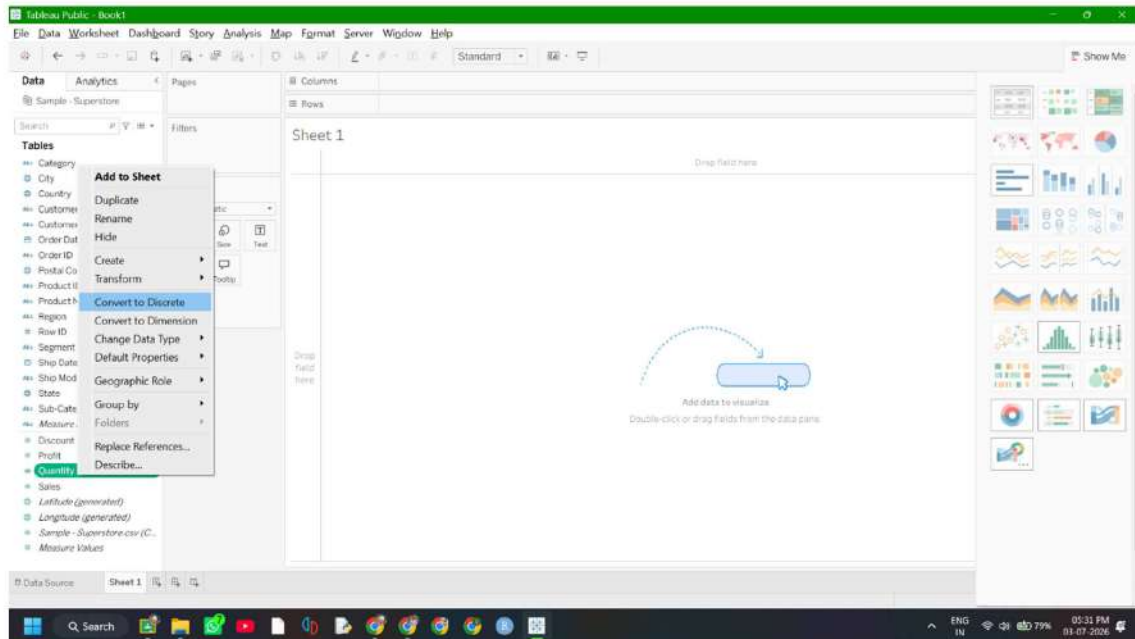
Tableau colors its data pills to represent how it visualizes information. **Blue means Discrete** (separate header blocks) and **Green means Continuous** (an infinite unbroken axis).

1. Find the Quantity field in your green measures list.

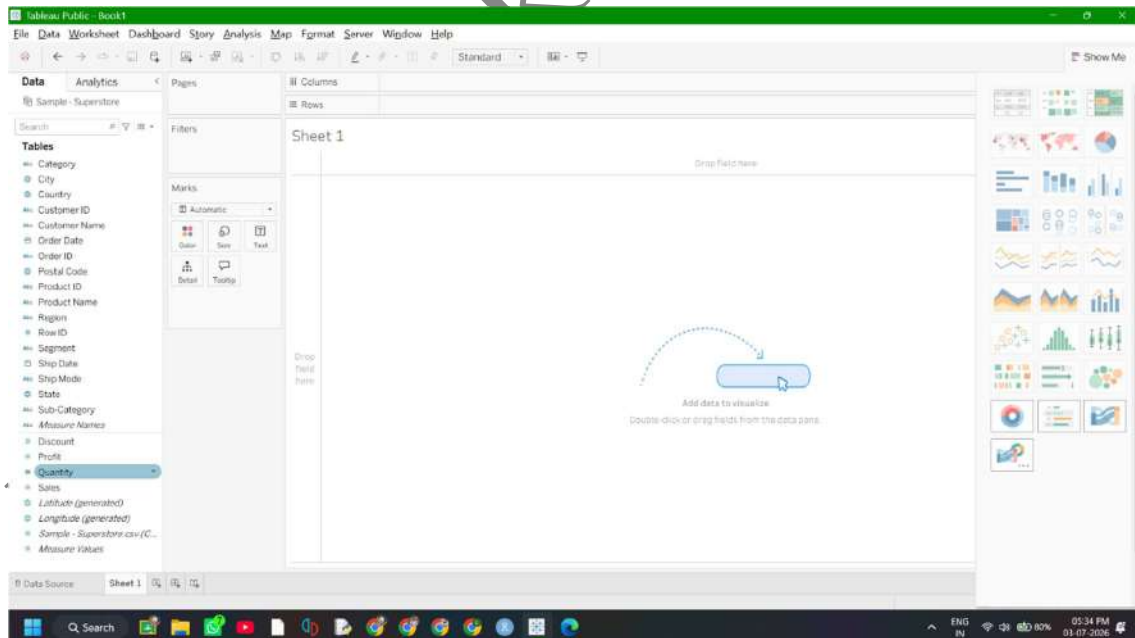


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2. Right-click (or hold Ctrl and click) on Quantity and choose Convert to Discrete. Notice its icon changes from a green to a blue.

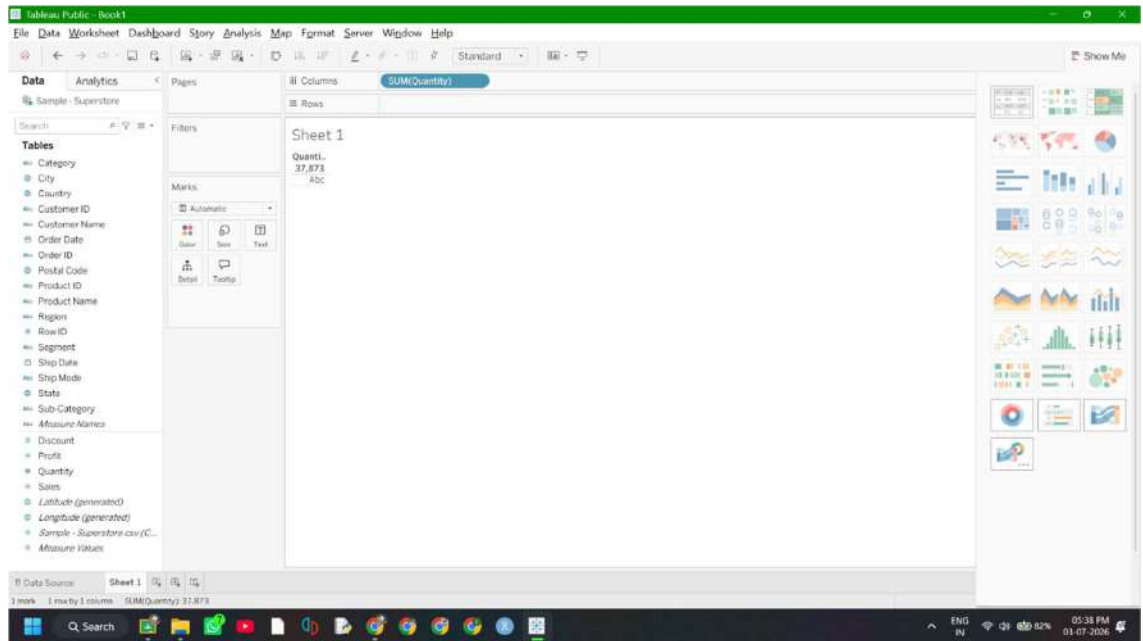


The Color changes from Green to blue

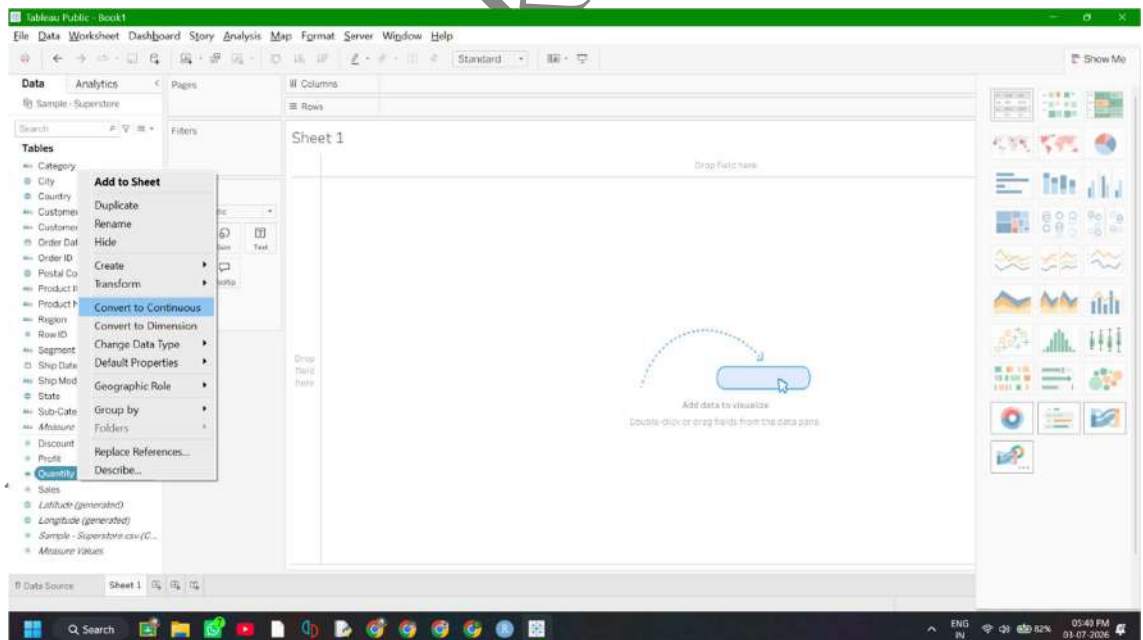


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3. Drag this new blue Quantity pill onto your **Columns** shelf. It will draw isolated individual header columns for each exact number (1, 2, 3, etc.).

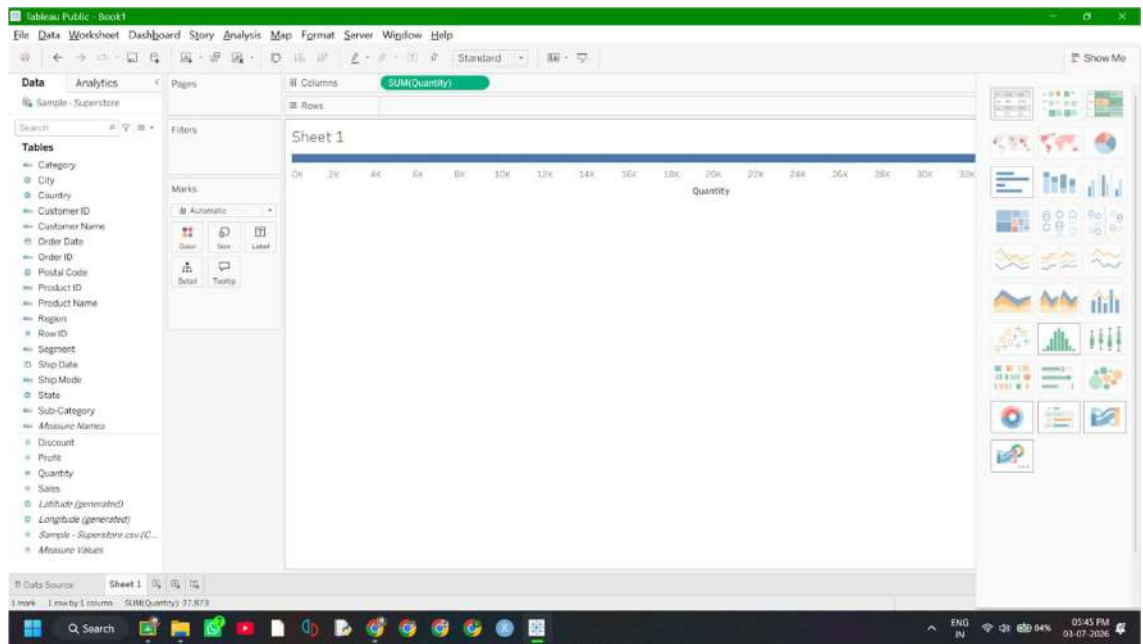


4. Drag it back off the shelf. Right-click Quantity again and select **Convert to Continuous**.



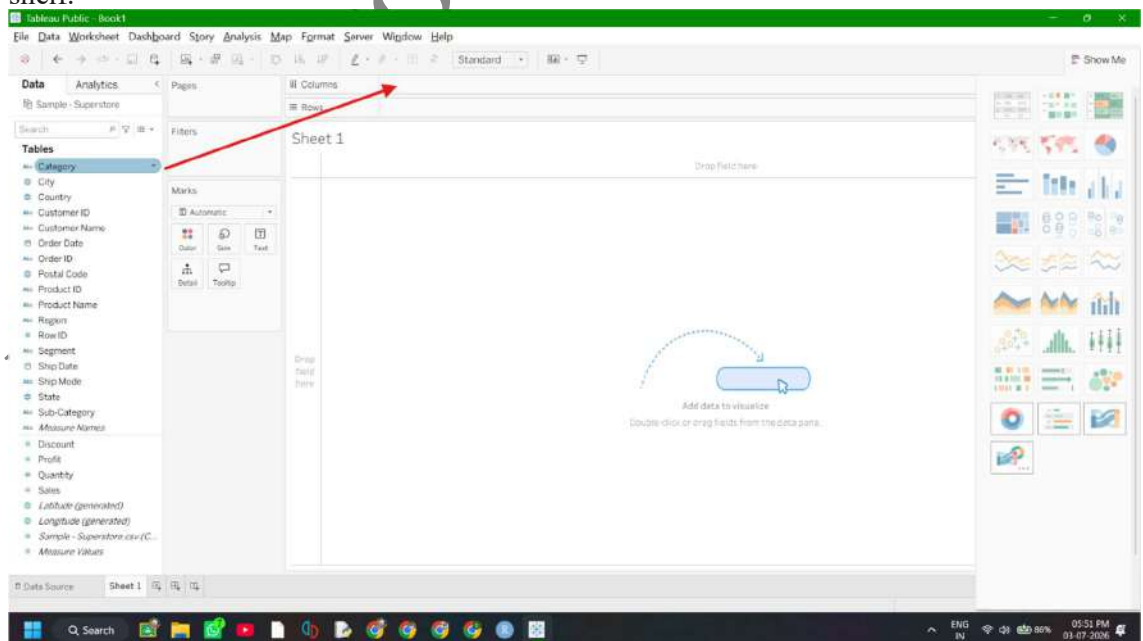
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5. Drag it back onto **Columns**. Instead of individual column blocks, Tableau now draws a singular, smooth, continuous numeric axis line.

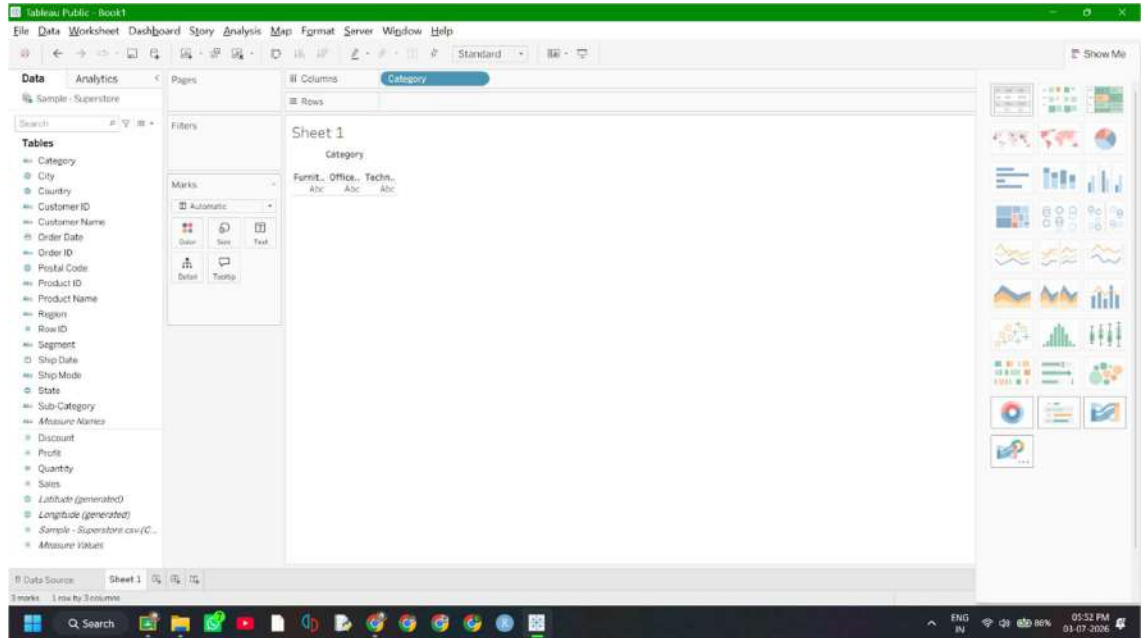


Part D: Create and Customize a Basic Bar Chart

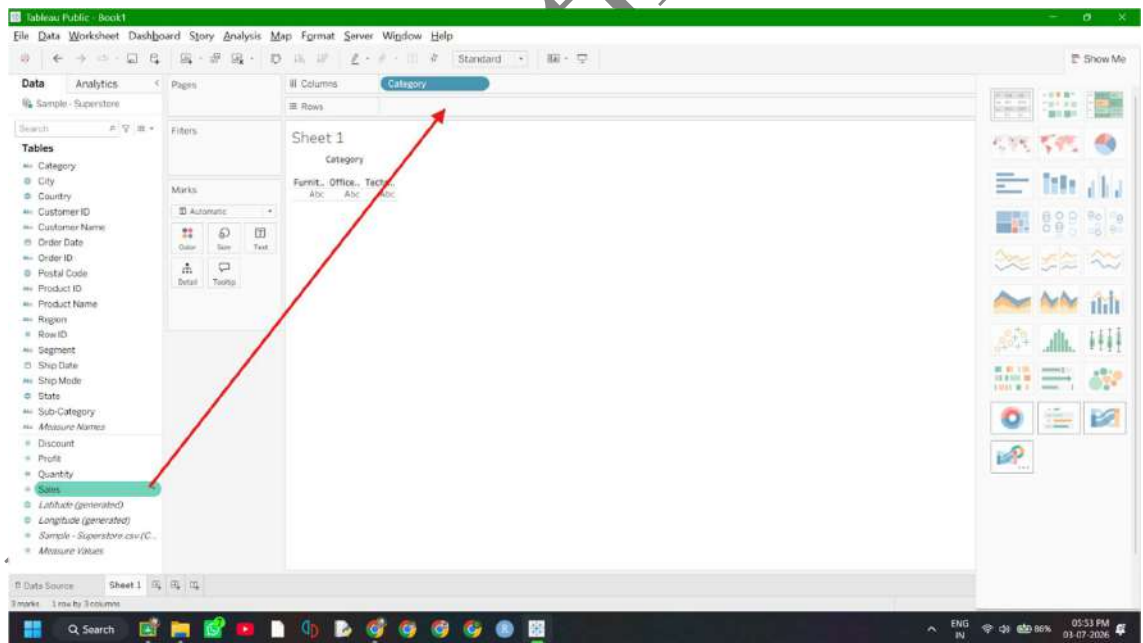
1. From your Data Pane, find the Category dimension and drag it up onto the **Columns** shelf.



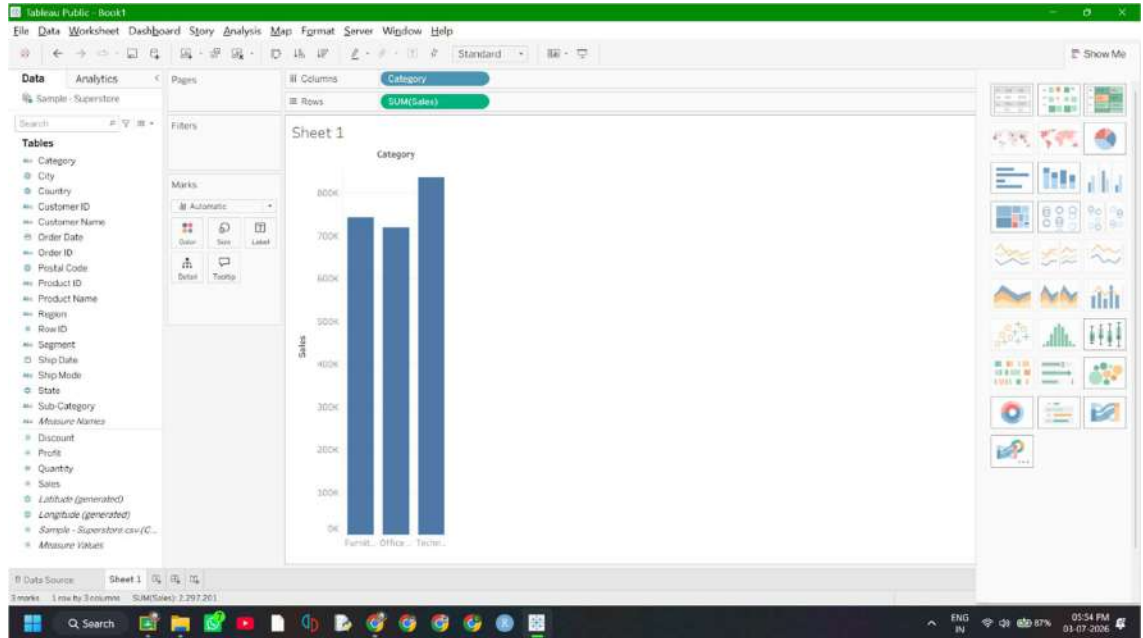
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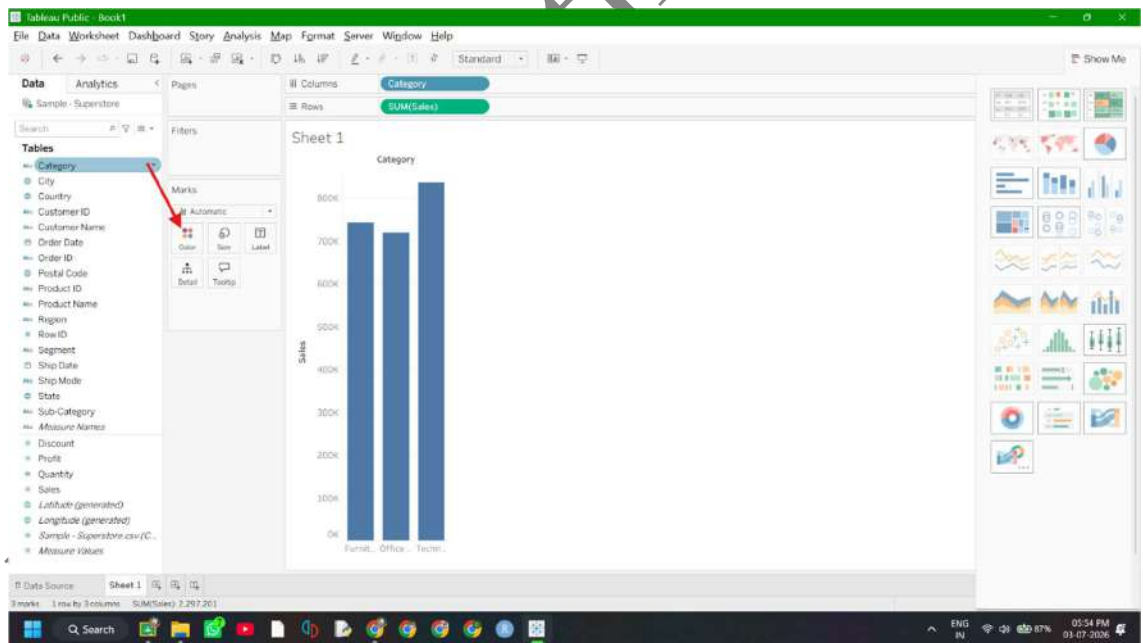
2. Find the Sales measure and drag it up onto the **Rows** shelf. You will instantly get three vertical bars representing Furniture, Office Supplies, and Technology.



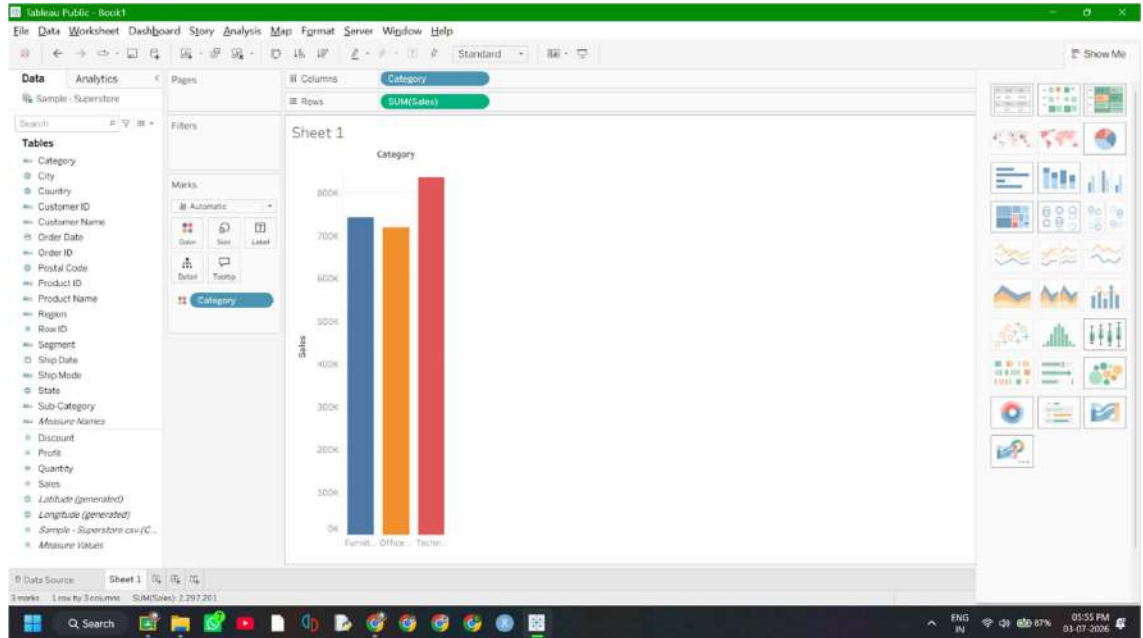
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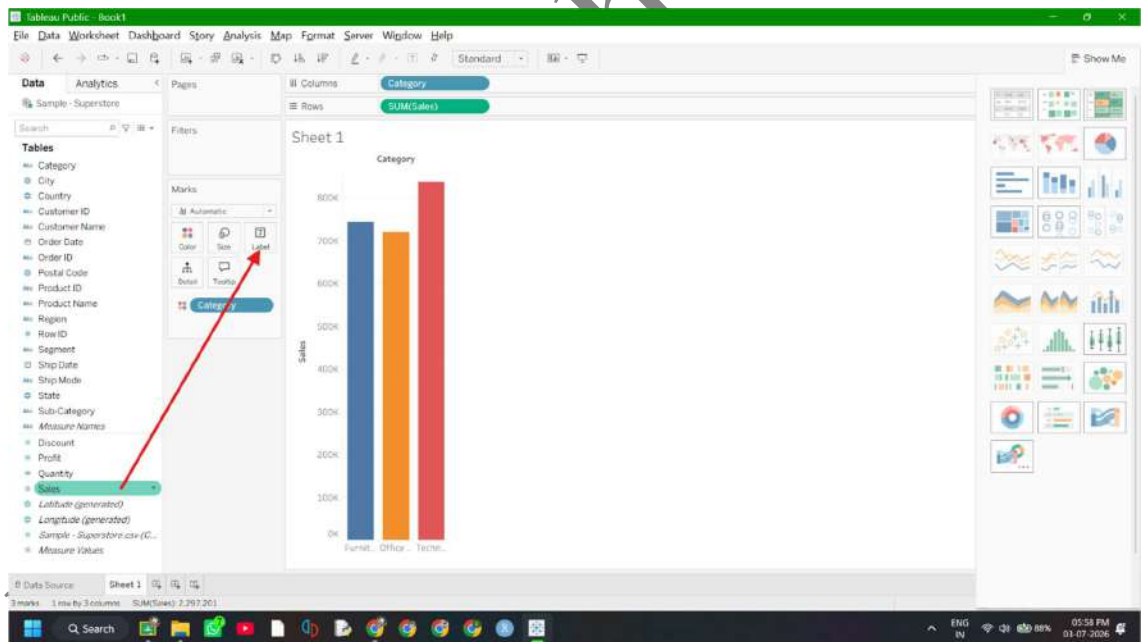
3. **Color customization:** Take Category from the Data pane a second time, and drop it directly onto the **Color** tile in the Marks card.



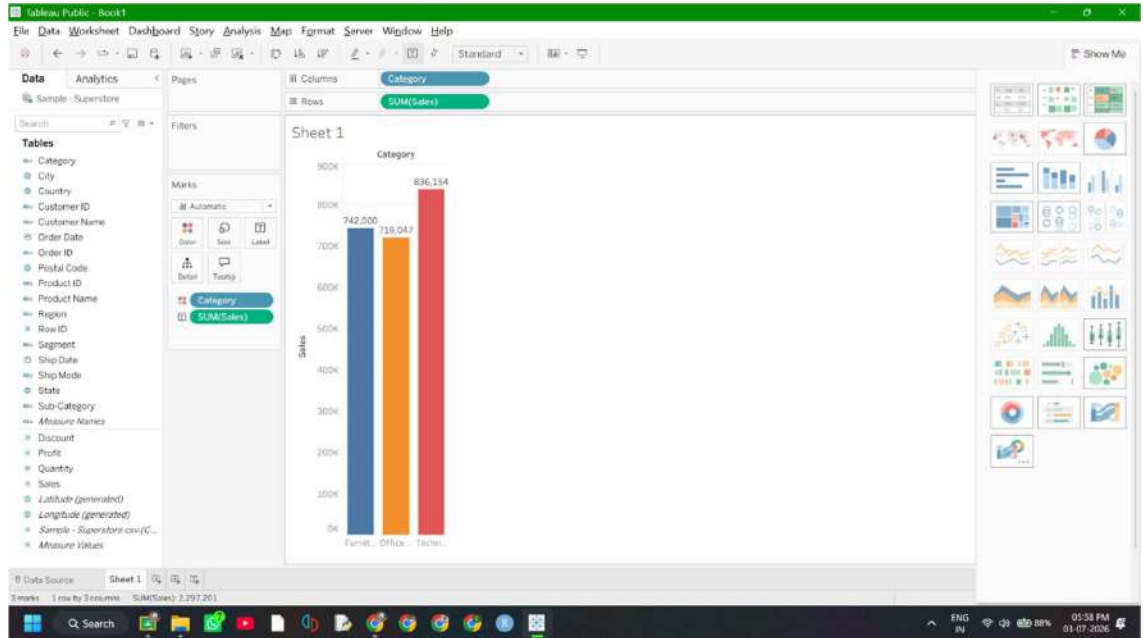
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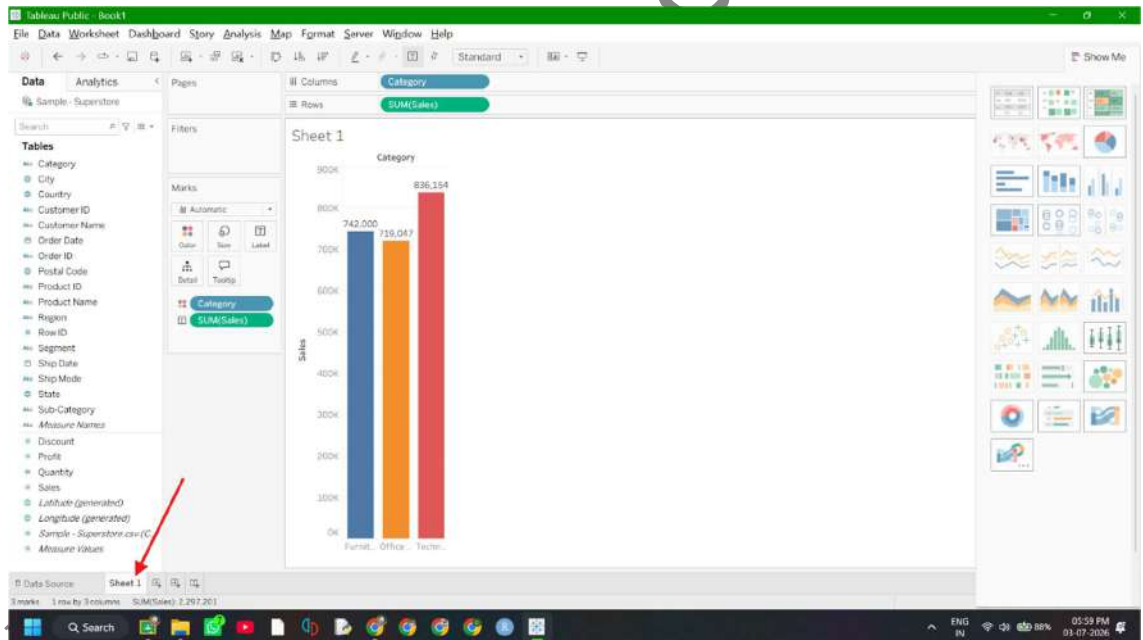
4. **Label customization:** Drag the Sales pill from the data pane and drop it onto the **Label** tile in the Marks card to display the precise revenue numbers over the bars.



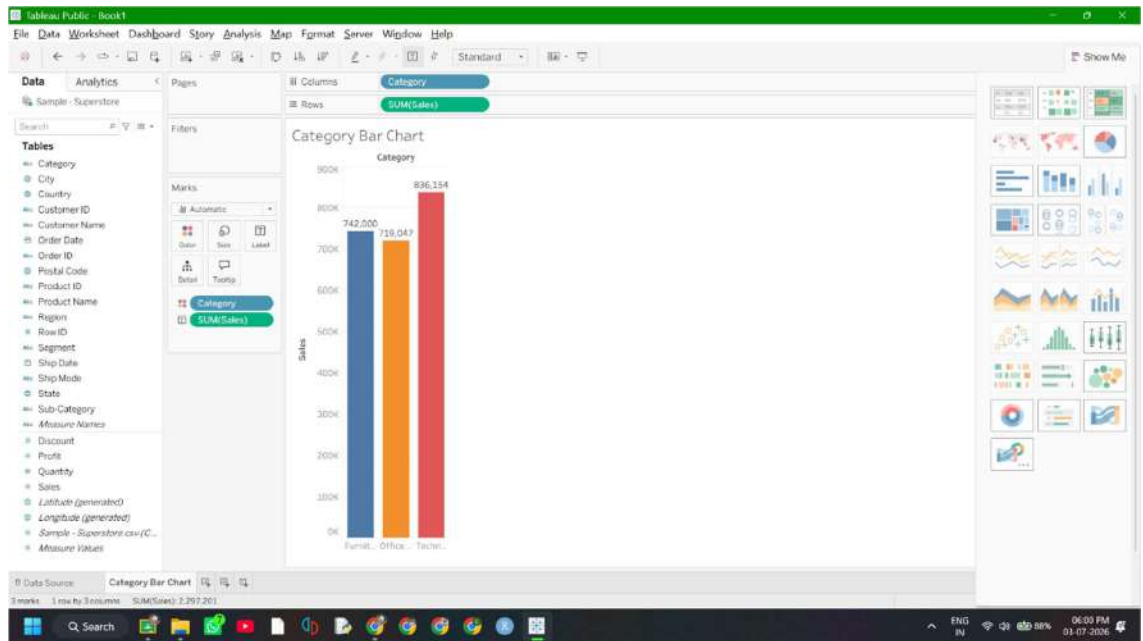
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5. Double-click the bottom tab sheet name Sheet 1 and rename it to **Category Bar Chart**.

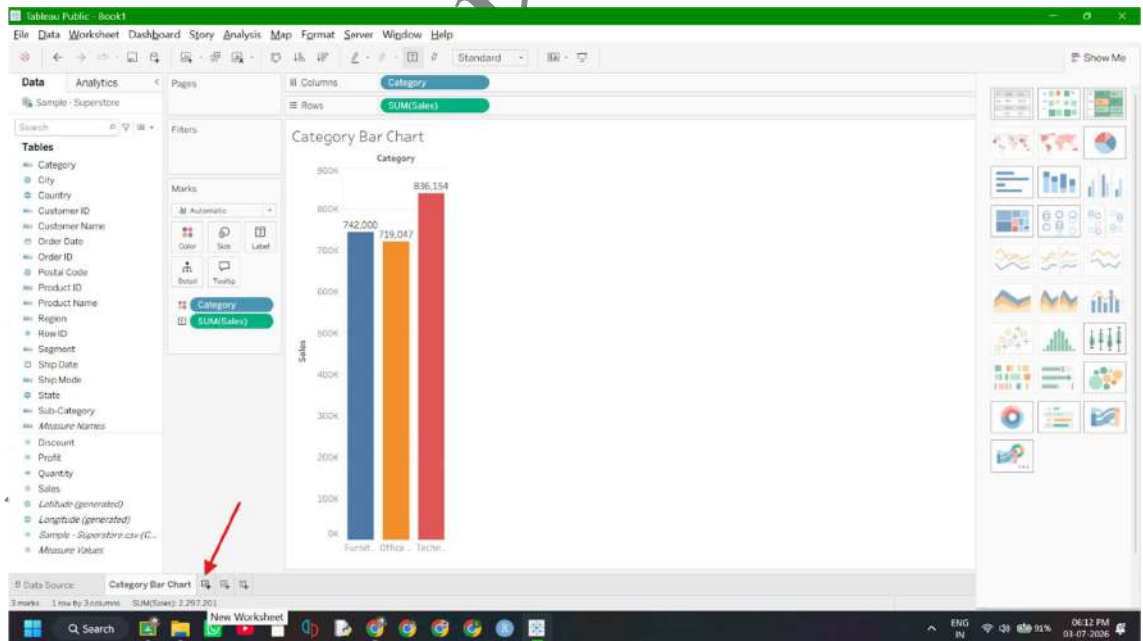


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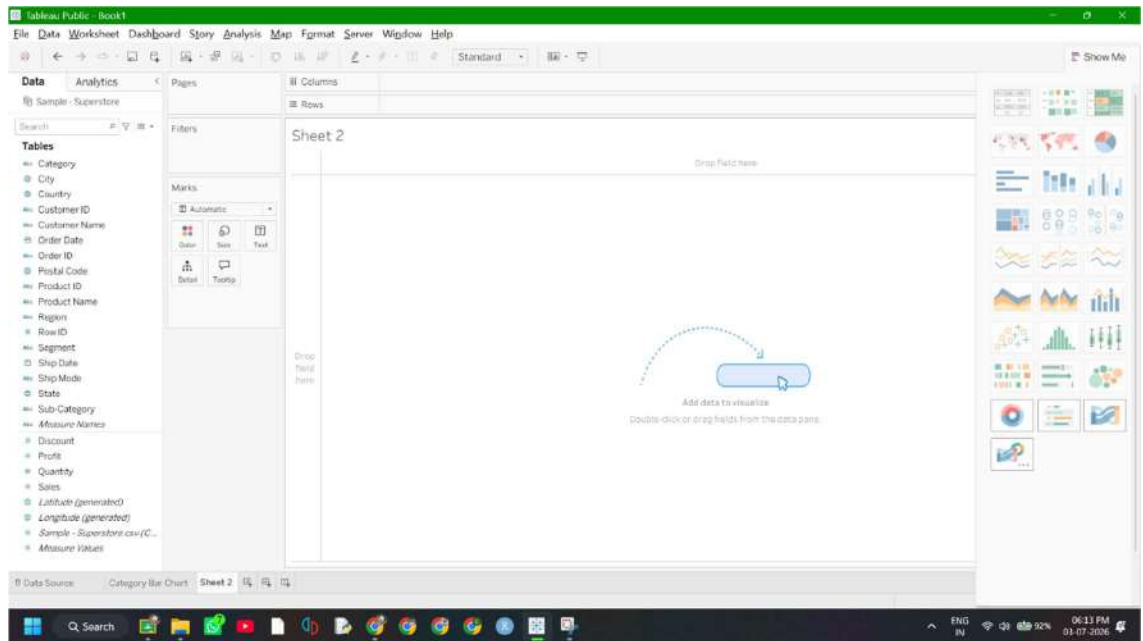


Part E: Build a Line Chart for Time-Series Analysis

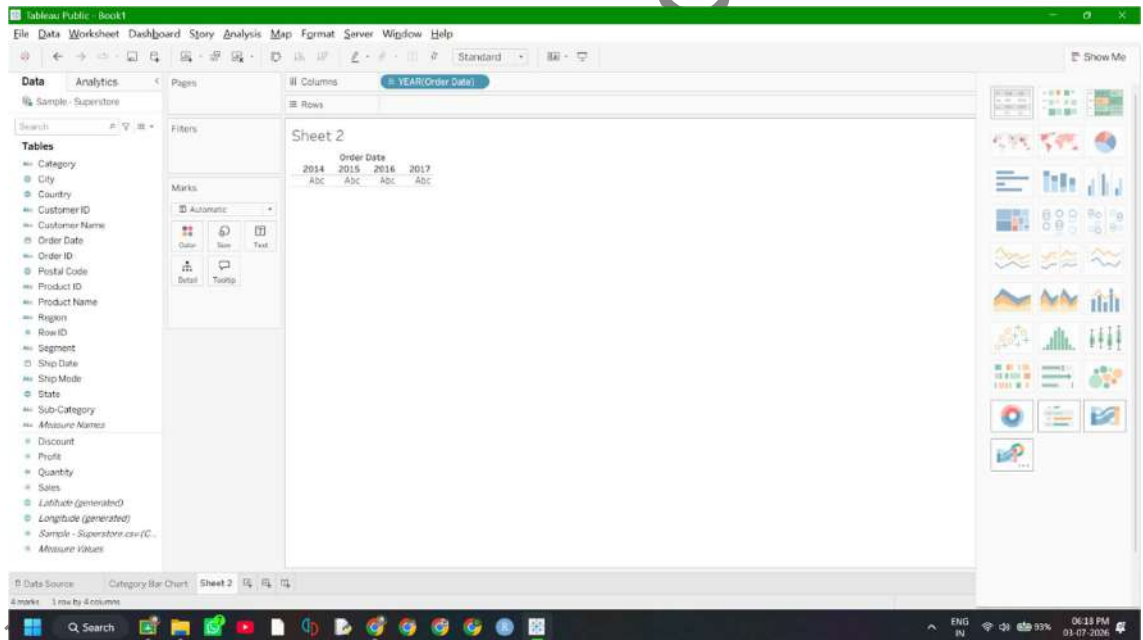
1. Click the **New Worksheet** button at the very bottom tab bar (the small icon with a chart and a plus symbol).



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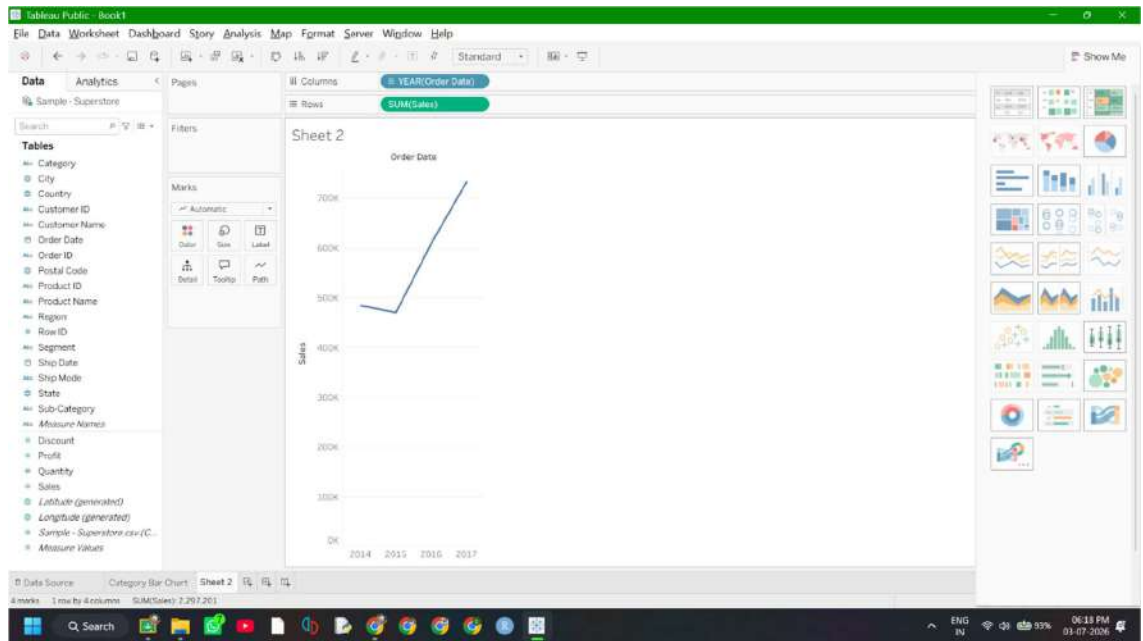


2. Drag the Order Date dimension onto your **Columns** shelf

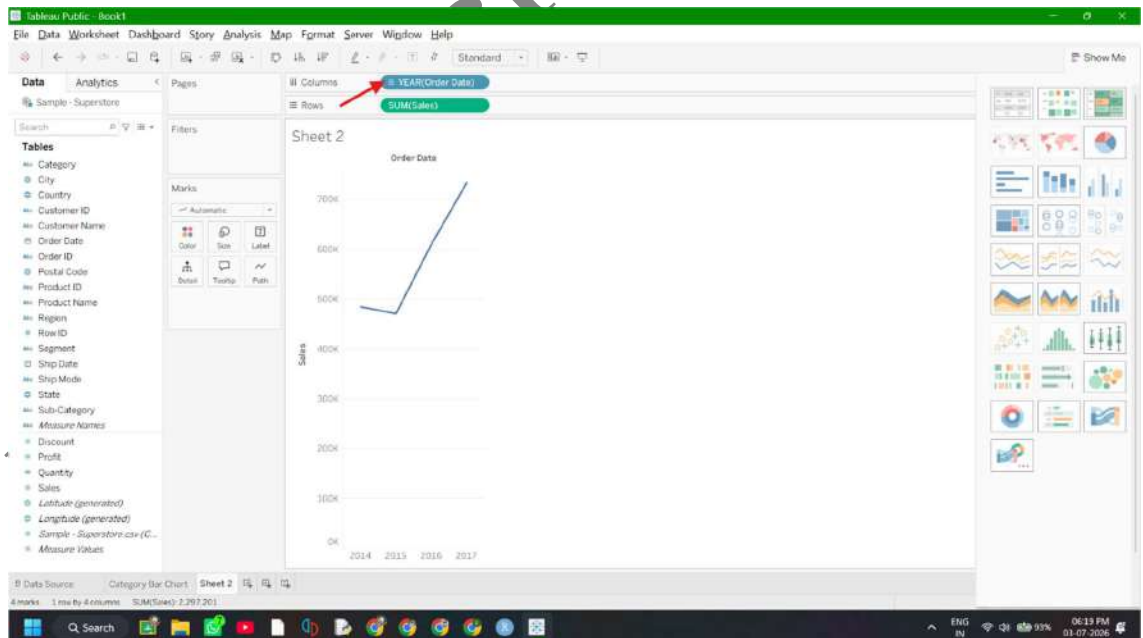


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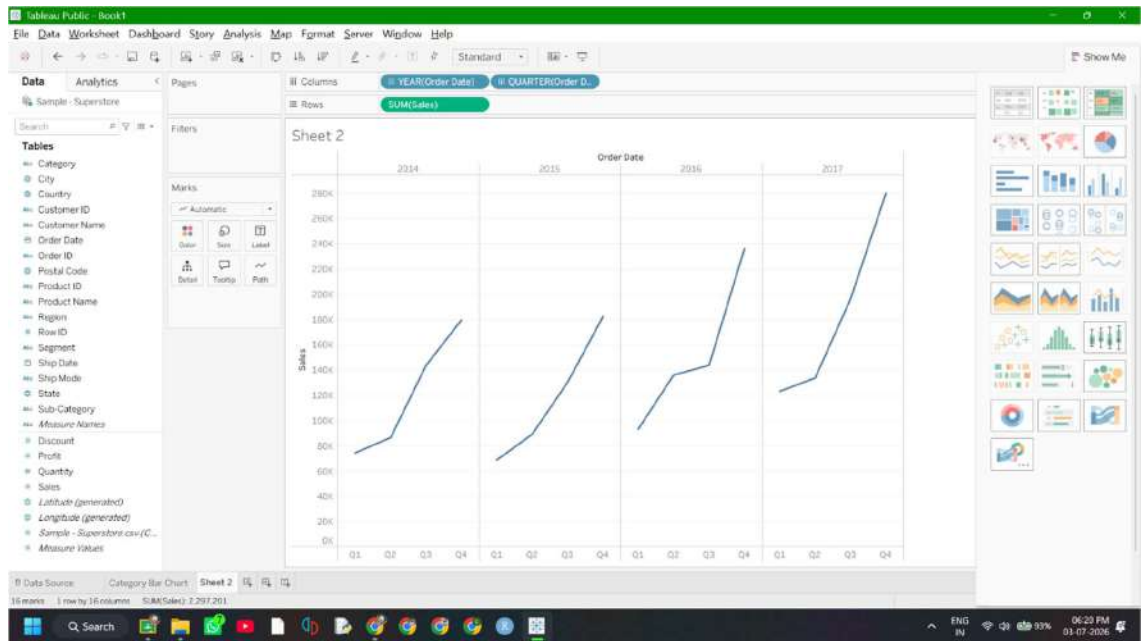
3. Drag the Sales measure onto your **Rows** shelf. Because a date field is on your columns shelf, Tableau defaults straight into a line chart.



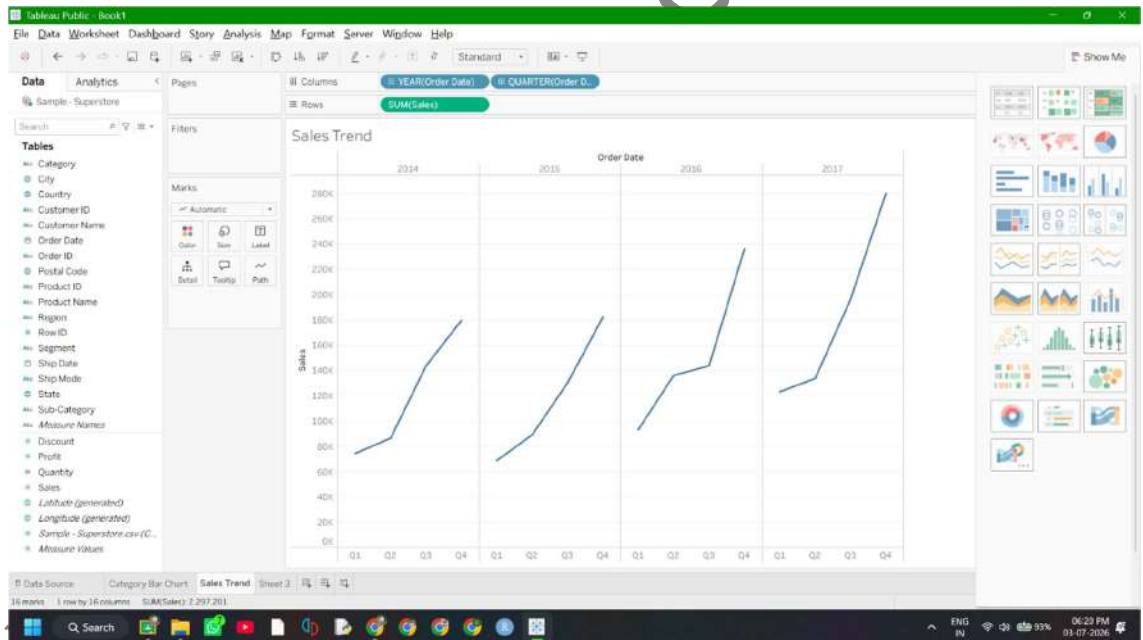
4. **Drilling Down:** Look at the blue YEAR(Order Date) pill sitting in your Columns shelf. Click the small + icon located on that blue pill. It will expand your data into Quarters and Months, revealing hidden sales trends over time.



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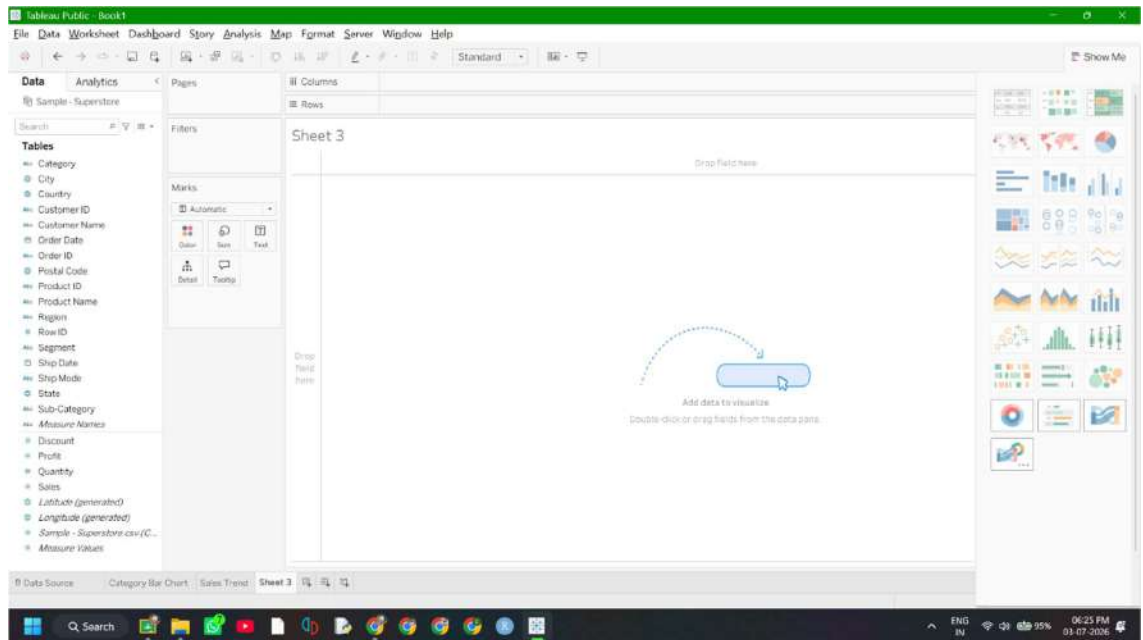


5. Double-click the sheet tab at the bottom and rename this worksheet to **Sales Trend**.

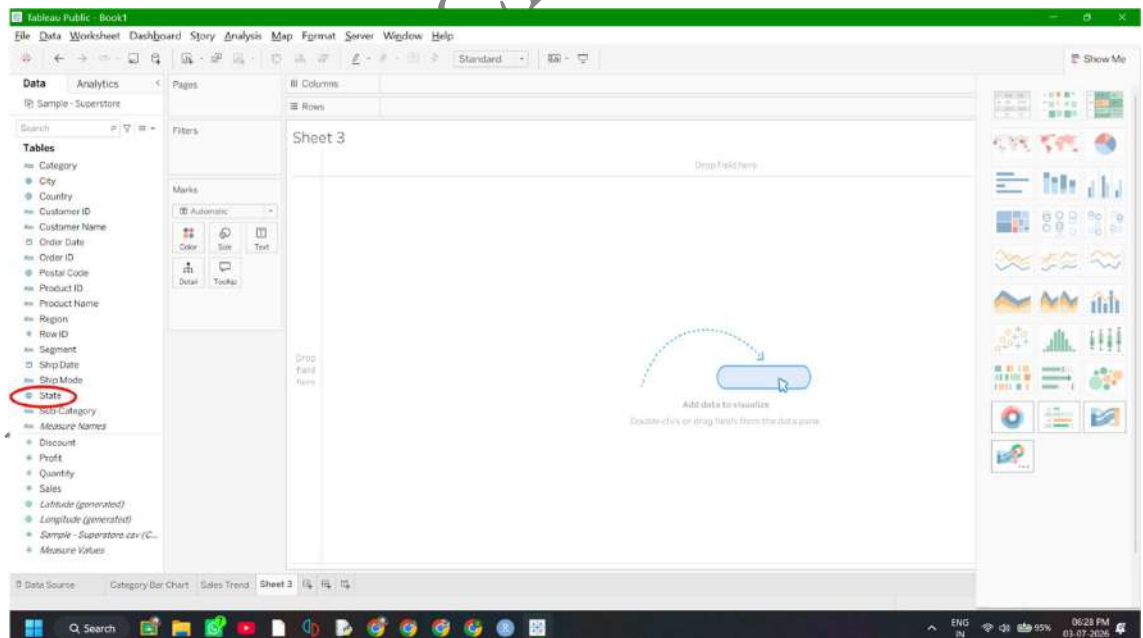


Part F: Create a Geographic Map Visualization

1. Click the **New Worksheet** button at the bottom to create your third sheet.

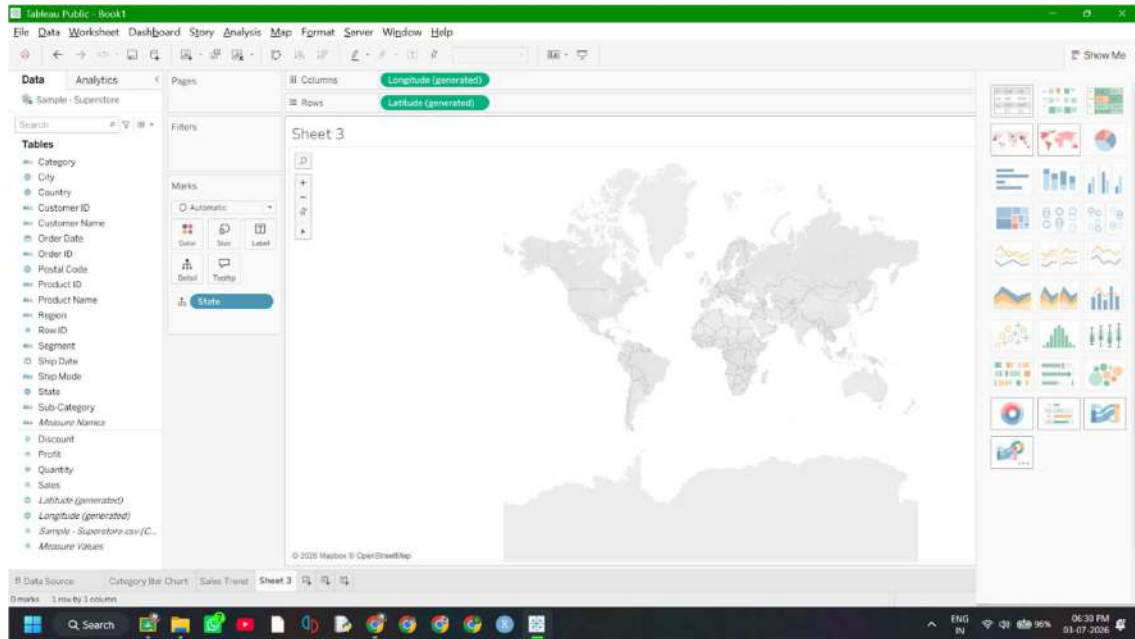


2. Look through your Data Pane and find the State field which is marked in red circle. It should display a tiny globe icon next to it, indicating geographic data.

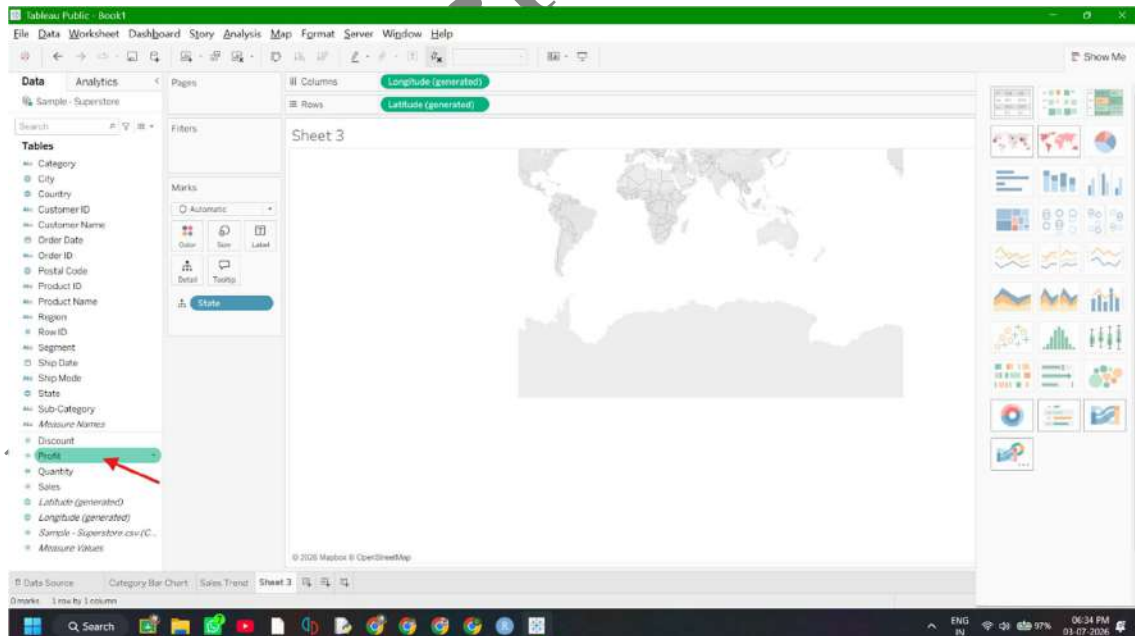


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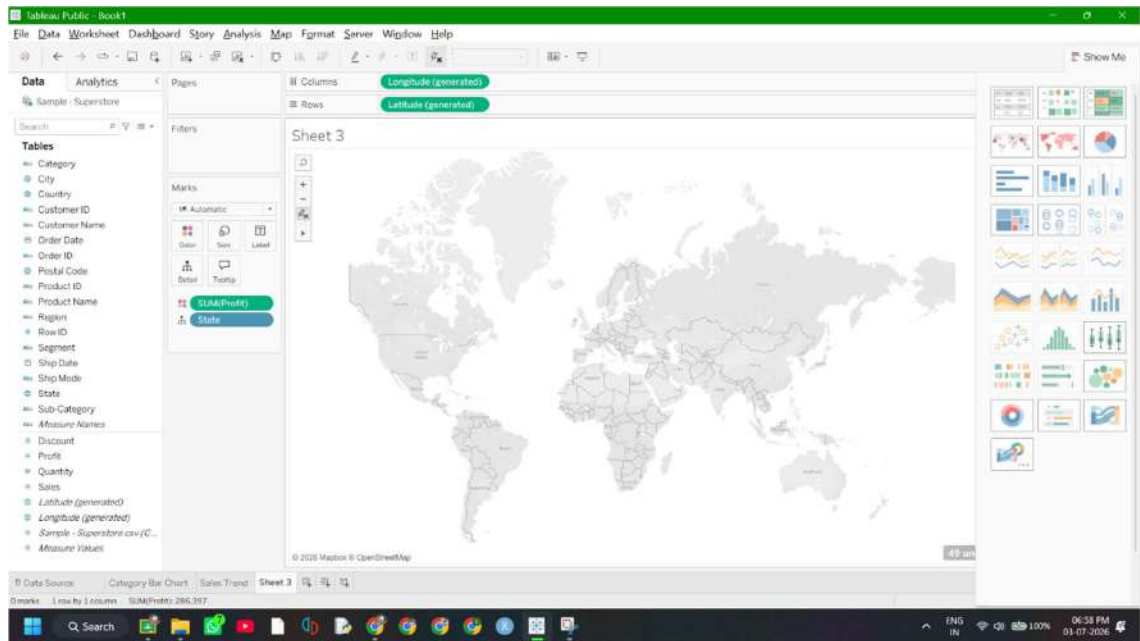
3. **Double-click** the State field. Tableau will automatically inject Latitude and Longitude coordinates into your shelves and draw a map.



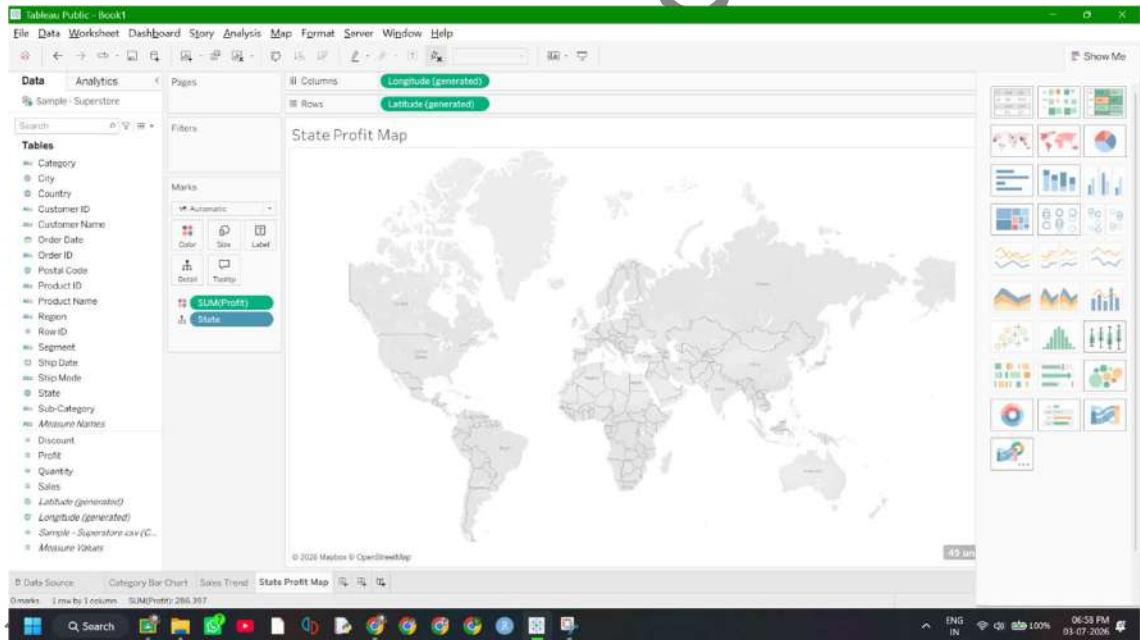
4. **Styling the Map:** Go to your Data Pane, pick up Profit, and drop it directly onto the **Color** icon inside your Marks card. This will instantly color-code each state based on its performance.



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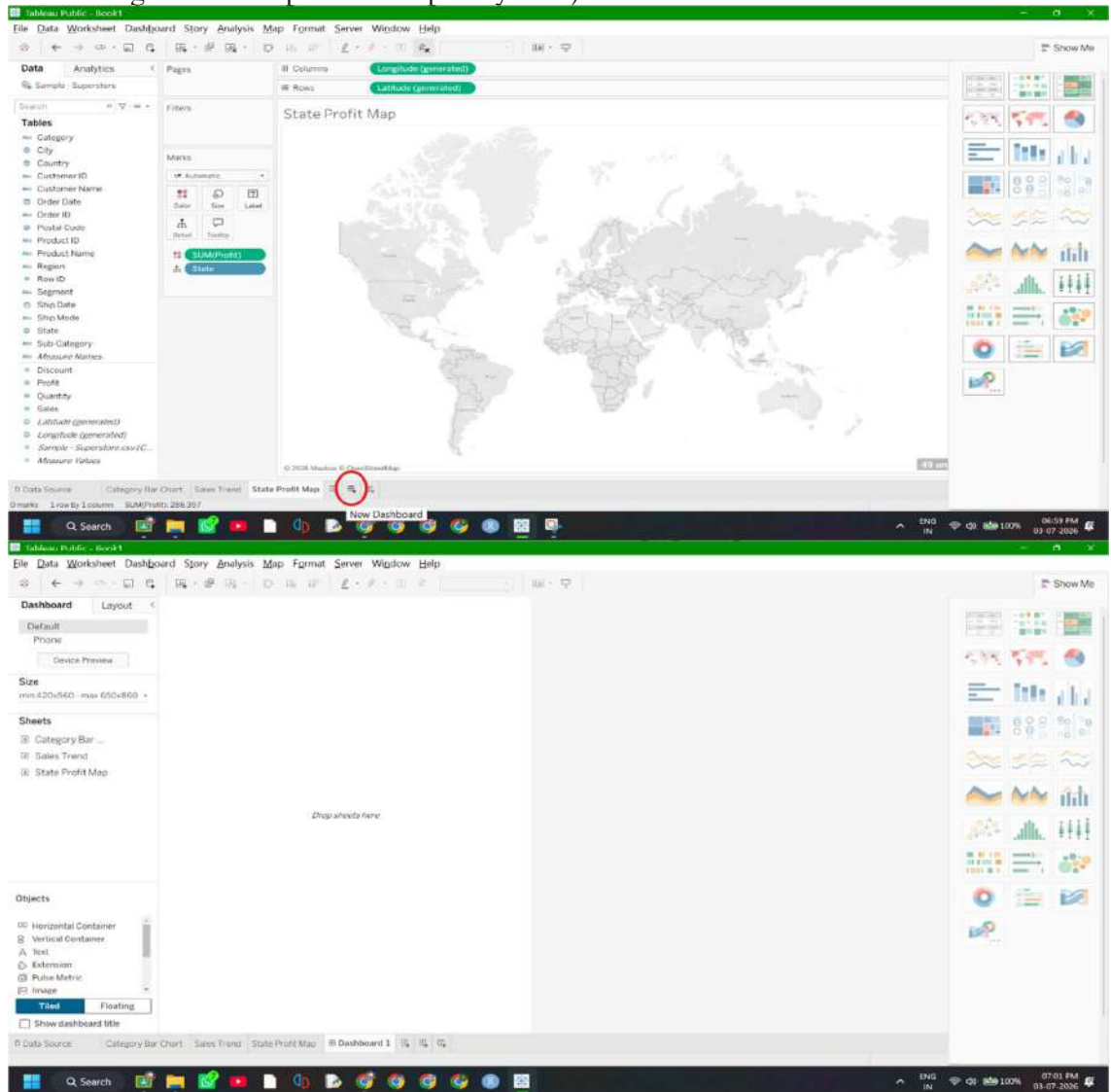


5. Rename this worksheet tab at the bottom to **State Profit Map**.



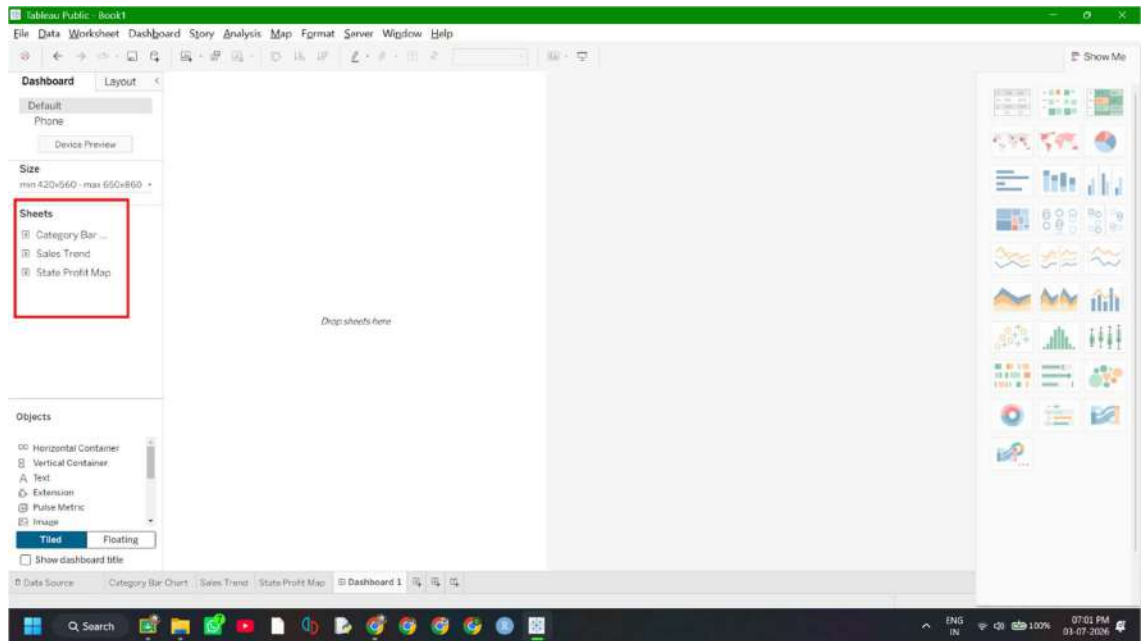
Part G: Combine Charts into an Interactive Dashboard

1. Look at the bottom navigation tab bar and click the **New Dashboard** icon (it looks like a small 2x2 grid window pane with a plus symbol).

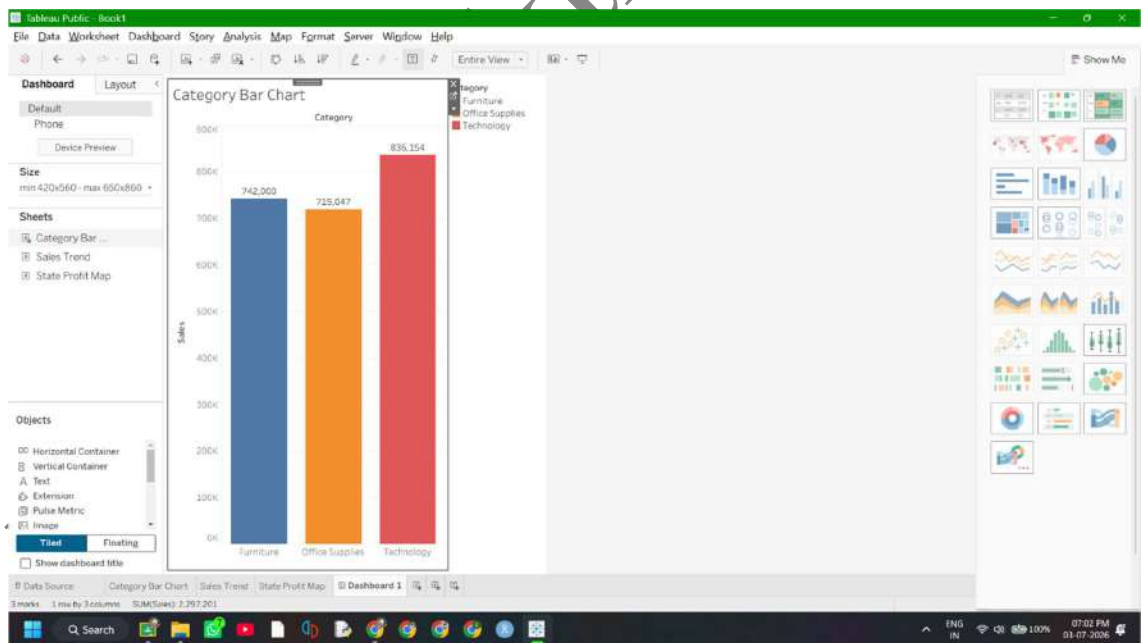


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2. On the left side panel, you will see a list of your worksheets: *Category Bar Chart*, *Sales Trend*, and *State Profit Map*.

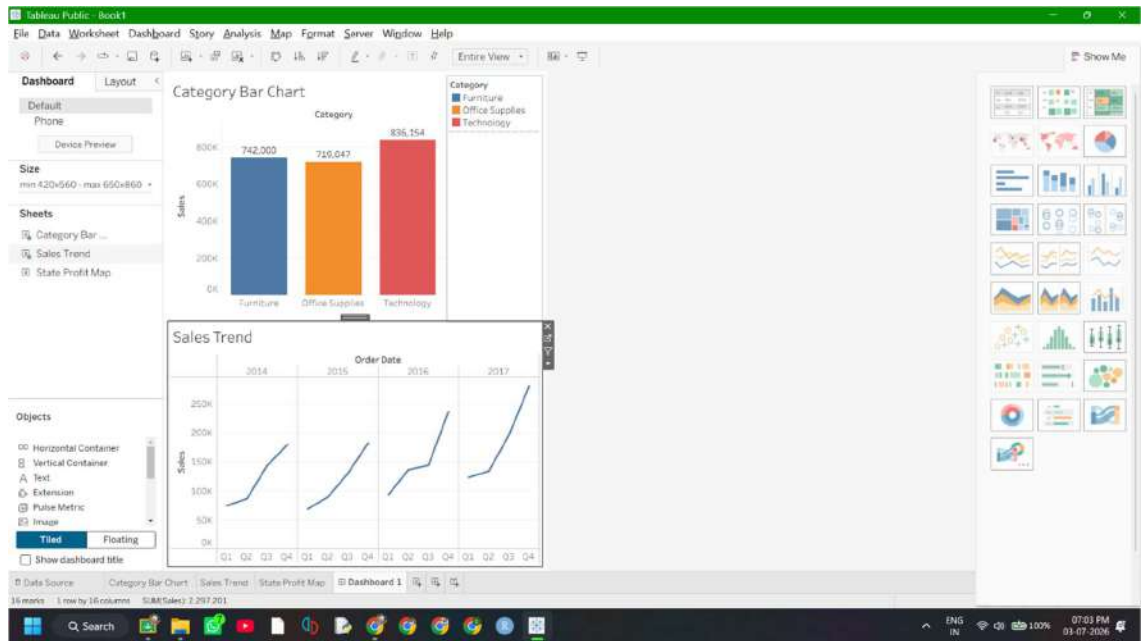


3. Drag your **Category Bar Chart** sheet onto the empty dashboard canvas area.

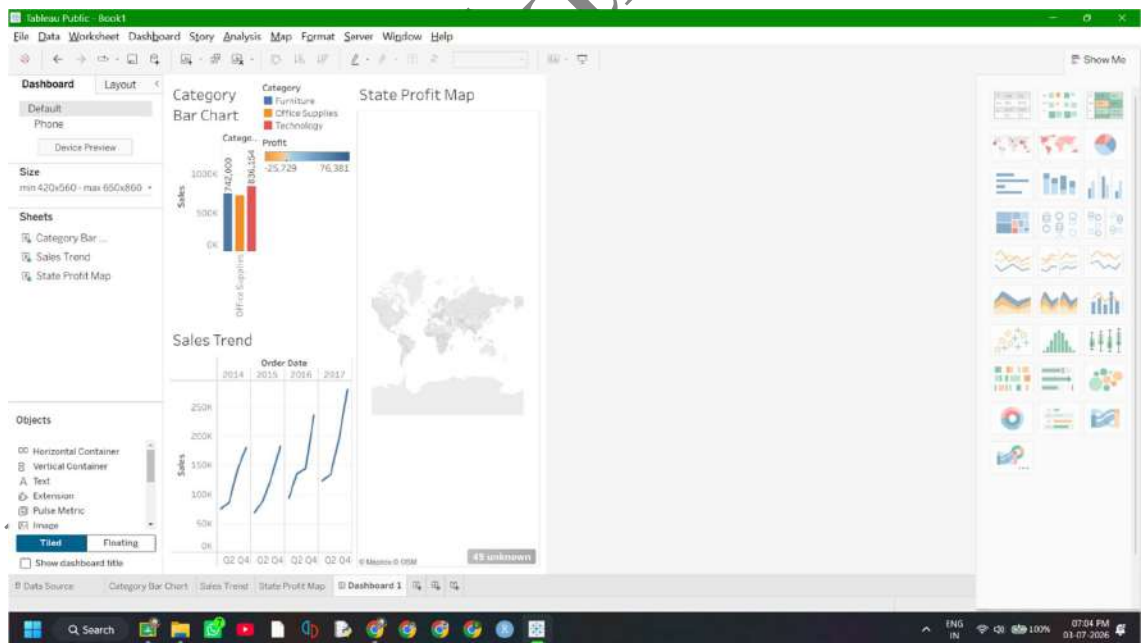


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4. Drag your **Sales Trend** sheet and drop it right along the bottom edge of the bar chart workspace.



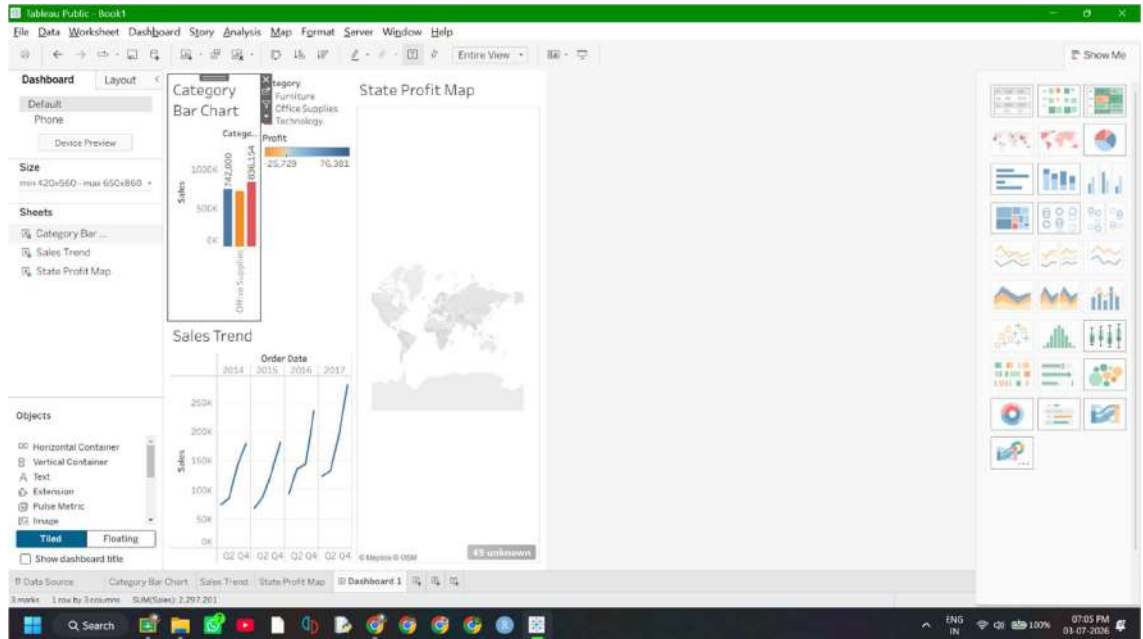
5. Drag your **State Profit Map** and drop it alongside the right side of the layout.



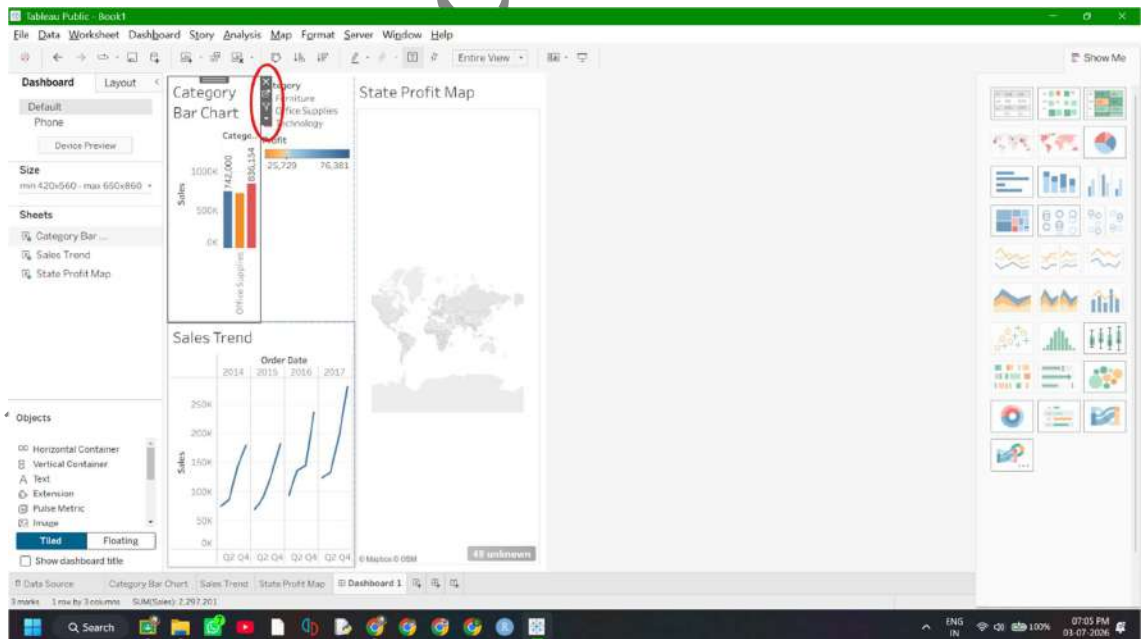
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Turning on the Magic: Interactivity

1. Click once inside your **Category Bar Chart** zone on the dashboard to select it.

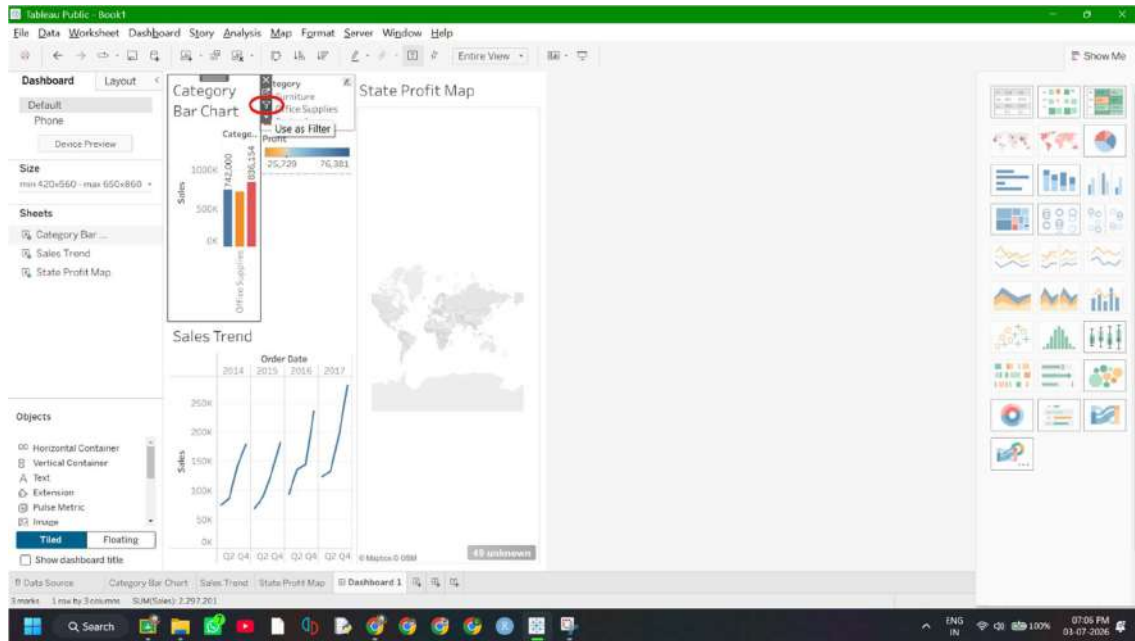


2. Look at the subtle gray menu options border that opens up on its top-right corner marked in red circle.



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3. Click the tiny **Funnel Icon (Use as Filter)**.



Now test out your creation! Click on the **Technology** bar in your bar chart. The map and the time-series line chart will dynamically isolate and update to display data *exclusively* for your Technology sales. Click the bar a second time to reset the filter.

